



Association of Zoos & Aquariums

2023 AZA CONSERVATION GRANTS FUND

Application Instructions

DEADLINE

This 2023 Conservation Grants Fund (CGF) electronic application must be used for all proposals. Completed applications must be submitted via email no later than **5:00 p.m. EDT** on **15 March 2023**. An auto-response email with a time-stamp will confirm receipt of the application.

*Incomplete applications, or applications received after 5:00 p.m. EDT on 15 March 2023, **regardless of cause**, will NOT be eligible for consideration. NO DEADLINE EXCEPTIONS WILL BE MADE.*

MEMBERSHIP

The Principal Investigator (PI) and all Co-Investigators (CIs) *must be individual AZA members* at the Professional Associate, Professional Affiliate, Professional Fellow, or Student level. Institutional membership does not count as individual membership and individual members at the Friend level are not eligible. Other participants actively involved in the completion of the project should be listed as collaborators and are not required to be AZA members. To apply for membership, visit <https://www.aza.org/individual-membership>.

FUNDING TIMELINE

All CGF projects must be completed within 12 months. CGF grants are intended to serve as seed money to attract additional funding and proposals should request funds only for this one-year period. Identical components of a project will not be funded for more than two grant cycles; however, funding requests for different components of the same project may be applied for in subsequent grant cycles.

CGF proposals undergo a two-tiered review process, outlined at <https://www.aza.org/cgf-selection-process>. Applicants will be notified of funding decisions in early September 2023 and funds will be available 2 October 2023. CGF funds can only be used for proposal expenses incurred on or after the award date. Expenses incurred prior to the award date will not be covered by CGF funds.

APPLICATION GUIDELINES

The application and all supplemental forms are posted online at: <http://www.aza.org/cgf-information-for-applicants/>. These documents contain form fields that must be completed and submitted via email to: CGFapp@aza.org. *All application materials must be in electronic format. No hard copies of any application materials will be accepted.*

It is strongly recommended that you save a copy of your completed application and supplemental forms as a backup. Only completed applications, submitted by the deadline defined above, will be reviewed. Applicants will be informed electronically if their application is incomplete.

APPLICATION INSTRUCTIONS AND CHECKLIST: All completed application materials must be submitted **in one email** by the deadline identified above and include the following:

Application: The application is one document consisting of the three sections outlined below. All sections must be complete in order for the application to be considered complete. The application must be submitted as a Microsoft Word document titled **"Application"** and attached to your Application email.

Applications submitted in Portable Document Format (PDF) will NOT be eligible for consideration. NO EXCEPTIONS WILL BE MADE.

Note that “spell checker” and “track changes” functions cannot be used in this form. AZA strongly recommends that applicants compose the content for their application in a separate document to make use of these features, and then copy and paste the final text into the form fields. When composing in a separate document, be aware of character limitations. If the copied text exceeds the character limit when pasted into a form field, *all text exceeding the character limitation will be cut off and will not appear in the final proposal.* Character limits include spaces.

Also note that certain formatting such as italics, bold, and underlining, is unavailable due to the application format. AZA recommends the use of numbering and spacing to emphasize important points in the text.

- ☒ **Cover Sheet.** Fill in the text fields for the project title, amount requested, and PI and CI information. Choose *two* project categories from the given list and identify the SAFE species that is (are) the focus of the project. Include a project abstract of 1500 or fewer characters.
- ☒ **Project Proposal.** Text fields are provided for responses to questions 1-12. Fields have a limit on the number of characters that may be used to keep the application concise. Character limits, including spaces, are stated following each question. If a question is not applicable to your proposal, write “N/A” in the field but *do not leave any form fields blank*, as this may delay processing of your application.
- ☒ **Budget.** Required budgetary information includes:
 - ☒ Detailed Project Budget. Detail how the budget will be allocated to specific items and indicate which funding source will pay for each item. State the total project budget, the amount requested from CGF, and other financial support applied for or received. Each of these amounts should reflect the one-year duration of the project. Funding decisions are announced in early September and CGF funds are to be used for work from the announcement date forward. Expenses incurred prior to the announcement will not be covered by CGF funds.
 - ☒ Budget Justification. Provide an explanation of why specific major items are essential to the project and reasonable in cost.

Required Attachments: The following must be attached to your Application email:

- ☒ **Curricula Vitae (CV).** A CV is required for each PI and CI listed on the Cover Sheet and must follow the format stated in these guidelines. CVs must not exceed two letter-sized (8.5 by 11 inch) pages each and must be typed in a legible font size (12 point recommended); each side of the paper counts as one page. CVs may be attached for collaborators but are not required. Each CV must be submitted as a separate document entitled, **“CV Investigator’s Last Name.”** CVs *exceeding the maximum length will be cut off at the second page.*
- ☒ **Statement of Institutional Support Forms.** Electronically signed *Statements of Institutional Support* from the CEO(s)/Institutional Director(s) (not the COO, General Curator, Vice President, Dean, or Department Director) of the PI’s institution and **each** CI institution *and* collaborating institution must be attached to the Application email as separate documents titled **“SIS Institution’s Name.”** The *Statement of Institutional Support* form can be found with the rest of the Application materials on the AZA website identified above.

Applicants wishing to include support letters from entities *other than* PI and CI institutions, collaborating institutions may submit such items as appendices.

Letters of support from PI and CI institutions and collaborating institutions will not be accepted in lieu of the *Statement of Institutional Support* form.

- ☒ **Statement of SAFE Species Program Alignment Forms.** Electronically signed *Statements of SAFE Species Program Alignment* from the program leader of relevant SAFE species program(s) must be attached to the Application email as separate documents titled “**SOA SAFE Species Program Name.**” The *Statement of SAFE Species Program Alignment* form can be found with the rest of the Application materials on the AZA website identified above.

Letters of support/endorsement from AZA Programs (Taxon Advisory Groups, Species Survival Plans®, SAFE) are **not** accepted, although coordination with AZA SAFE Programs is expected and should be described in Question 1. To assure program review of relevant submissions, AZA SAFE or Animal Program Leaders may serve as expert reviewers in the application review process and may express their support for a proposal at that time.

- ☒ **Permits.** If permits are required to complete your proposed project, you must attach a copy of each to your Application email. Each permit attached to the Application email must be titled “**Permit #.**” *If you are not able to obtain a required permit before submitting this proposal, you must submit it by 15 June 2023 to ensure award eligibility.*
- ☐ **Proposal Resubmission Form.** If a proposal for this project was previously submitted for AZA grant funding (i.e., CGF or SAFE granting program) and rejected, the revised proposal *must* include a completed *Proposal Resubmission Form* as an attachment on the Application email. This attachment must be submitted as a Microsoft Word document titled “**Resubmission Form**” and attached to your Application email. *Resubmission forms submitted in Portable Document Format (PDF) will not be accepted.* The form can be found with the rest of the Application materials on the AZA website identified above.

Note: Proposals that fail to provide all items identified in the above checklist in the correct format will not be reviewed.

Non-Required Attachments: The following attachments to your Application email are optional:

- ☒ **Citations (recommended).** Citations/references for statements made in the Application should be included in a separate document titled “**Citations**” and attached to the Application email.
- ☒ **Appendices.** Appendices are allowed, but may only *clarify*, and not *supplement*, information in the narrative. For example, photos or diagrams would be acceptable, but not videos, publications, or SAFE species program plans. Appendices must be attached to the Application email titled “**Appendix #.**”

Note: Appendices that do not adhere to these guidelines will be removed from the proposal prior to review.

SUBMISSION INSTRUCTIONS

By submitting your application, you agree, if funded, to abide by the Grant Terms and Conditions posted on AZA’s website at <http://www.aza.org/cgf-information-for-applicants/>.

Proposals must be submitted electronically using the 2023 Application and related forms via a **single email, with the proposal title as the subject line**, to CGFapp@aza.org no later than 5:00 p.m. EDT on 15 March 2023. Hard copies of any application materials or applications received after this deadline will not be accepted regardless of cause.

AZA's email system accepts messages up to 8 MB in size. In cases when applications are too large to be accepted in one email, AZA will accept the application submission in two separate emails, provided that the same proposal title is identified in the subject line of each.

The 2023 Application email **must** include the following attachments

- Application (this completed form);
- CVs for each Principal and Co- Investigator(s);
- Statements of Institutional Support for each PI and CI institution and collaborating institution;
- Statement(s) of SAFE Species Program Support; and
- A Proposal Resubmission form, if applicable.

The Application email may also include citation and appendices attachments.

Auto-response emails will be sent to applicants confirming receipt of the application package. If an auto-response is not received, please resubmit the application. If an applicant submits more than one proposal, the person should ensure that an auto-response was received for *each* submission. If submission problems arise, contact the AZA office at 301-562-0777. If you submit your application after the 5:00 p.m. deadline, you may still receive an auto-response from the Application inbox; however, your proposal will not be reviewed.

Applications which fail to follow submission instructions or are time-stamped after the deadline will not be considered and applicants will be notified that their application is inadmissible.

APPLICATION ASSISTANCE

The CGF website (<http://www.aza.org/cgf-information-for-applicants/>) contains a wealth of information to help applicants compose a competitive proposal.

Proposals received before 8 March will be reviewed by AZA staff to ensure that the application is completed correctly. Early submission allows time for AZA staff to inform applicants if problems are noted in time for resubmission. Please note that AZA staff does not review early proposals for content.

2023 AZA CONSERVATION GRANTS FUND Cover Sheet

Project Number:
For AZA use only

Title of Project: Understanding the ecology of tularemia in endangered in-situ black-footed ferrets and the prairie dog ecosystem

Amount Requested: \$18,385 (round to the nearest whole dollar)

Project Start Date: 11/1/2023

End Date: 10/31/2024

(CGF funding is for one year only, beginning no earlier than the award date)

Investigators: Each Principal Investigator (PI) and Co-Investigators (CI) *must be an individual AZA member at the time of application* at the Professional Associate, Professional Affiliate, Professional Fellow, or Student level. Other participants should be listed as collaborators on this form.

A Curriculum Vitae (CV) is required for each PI and CI, and a Statement of Institutional Support (SIS) form is required for each PI and CI Organization.

1. Principal Investigator (PI) Name and Title: Travis Livieri, Executive Director and Biologist

CV Attached? ☒ (required – no more than 2 pages)

Checking this box indicates agreement to the CGF Terms and Conditions, including affirmation that the organization is or will be a program partner of the relevant SAFE species program(s) upon acceptance of an award (required; available at: <http://www.aza.org/cgf-information-for-applicants/>). ☒

Individual AZA Member Number: 0810297461 (required)

Institution/Organization: Prairie Wildlife Research

Statement of Institutional Support Form Attached? ☒ (required)

Institutional Address: PO Box 643, Stevens Point, WI 54481

Phone: 970-219-1659

E-mail: tlivieri@prairiewildlife.org

2. Co-Investigator (CI) Name and Title: Shelli Dubay, Professor of Wildlife

CV Attached? ☒ (required – no more than 2 pages)

Checking this box indicates agreement to the CGF Terms and Conditions, including affirmation that the organization is or will be a program partner of the relevant SAFE species program(s) upon acceptance of an award (available at: <http://www.aza.org/cgf-information-for-applicants/>). ☒ (required)

Individual AZA Member Number: 0812265651 (required)

Institution/Organization: University of Wisconsin - Stevens Point

Statement of Institutional Support Form Attached? ☒ (required)

Institutional Address: College of Natural Resources, 800 Reserve St., Stevens Point, WI 54481

Phone: 715-346-4178

E-mail: sdubay@uwsp.edu

3. Co-Investigator (CI) Name and Title: Madisen Hartlaub, Wildlife Technician

CV Attached? ☒ (required – no more than 2 pages)

Checking this box indicates agreement to the CGF Terms and Conditions, including affirmation that the organization is or will be a program partner of the relevant SAFE species program(s) upon acceptance of an award (available at: <http://www.aza.org/cgf-information-for-applicants/>). ☒ (required)

Individual AZA Member Number: 081226446 (required)

Institution/Organization: Prairie Wildlife Research

Statement of Institutional Support Form Attached? ☒ (required)

Institutional Address: PO Box 643, Stevens Point, WI 54481

Phone: 920-619-9255

E-mail: mhartlaub@prairiewildlife.org

Information about additional co-investigators may be included in a separate document titled “**Additional Co-Investigators**” and attached to your Application email.

Collaborating Institutions or Organizations: A collaborating institution is defined as any institution actively involved in the design or completion of the project. Funding institutions *not* actively involved in project design or completion should be listed in the “Grant/Matching Support” section of the Project Budget and need not be listed as collaborators.

List all collaborating institutions below, including the institutions of the PI and CIs listed on the Cover Sheet, as well as those associated with the personnel listed in Question 6, and attach a Statement of Institutional Support form for each collaborating institution to the Application email.

Collaborating Institution Names:

UW-Stevens Point *Statement of Institutional Support Form Attached?* ☒ (required)

Prairie Wildlife Research *Statement of Institutional Support Form Attached?* ☒ (required)

Statement of Institutional Support Form Attached? ☐ (required)

Statement of Institutional Support Form Attached? ☐ (required)

Statement of Institutional Support Form Attached? ☐ (required)

Note: If additional Collaborating Institutions or Organizations need to be included, add their information in a separate document titled “**Additional Collaborators**” and include it as an attachment to your Application email.

Project Categories (choose two that best describe the project):

☐ Animal / Population Management (*Ex situ*)

☐ Conservation Education / Engagement

☐ Animal Health

☒ Field Conservation (*In situ*)

☐ Animal Welfare

☒ Research

☐ Communications / Public Relations

Identify the SAFE species program(s) this project will benefit: Black-footed ferret

Project Abstract (limit 1,500 characters with spaces):

Conata Basin/Badlands National Park in southwestern South Dakota is home to the largest wild population of endangered black-footed ferrets (*Mustela nigripes*; BFF) in the world. Diseases such as plague and canine distemper virus have caused dramatic decreases in BFFs and their primary prey, prairie dogs (*Cynomys* spp.), however, tularemia also infects BFFs and causes mortality in prairie dogs. Caused by *Francisella tularensis*, tularemia is a highly infectious bacterial disease and transmission occurs through biting arthropods such as ticks, exposure to infected animal tissue, and contaminated food or water. Little is known about how tularemia is transmitted in the prairie ecosystem. Additionally, ticks and fleas have been collected from BFFs, but we do not understand the tick source (e.g., burrows, environment). Prairie Wildlife Research has been studying BFFs in South Dakota since 2001 and in 2015 found that 40% of the BFFs in all sex and age classes were exposed to tularemia. The goals of this project are to 1) assess seroprevalence of tularemia in wild BFFs across time, 2) identify locations of questing ticks in the environment, within prairie dog burrows and on the surface of prairie dog colonies and sample ticks parasitizing BFFs, and 3) understand epizootiology of tularemia in the prairie ecosystem. Results will increase knowledge on disease dynamics of tularemia within the grassland burrow ecosystem and its impact on BFF populations.

Project Proposal

1. Describe the overall project and its significance for conservation and/or management. What species, sites, habitats, populations, or questions are targeted or addressed and what critical needs will be met? Explain the project's relevance to the SAFE species' program team's priorities and how it supports implementation of the team's 3-year program plan and existing regional and/or global recovery plans. **LIMIT 5,000 CHARACTERS with spaces**

Emerging or re-emerging infectious diseases have recently increased and pose significant threats to many wildlife species (Tompkins et al. 2015), particularly endangered species (Smith et al. 2009). Climate change may exacerbate potential disease outbreaks (Rohr and Cohen 2020) and promote vectors (Ostfeld 2009), thereby adding further complications to endangered species conservation. For the black-footed ferret (BFF; *Mustela nigripes*), an endangered species with a global in-situ population of ~400 that is sensitive to several diseases and is reliant upon prairie dogs (*Cynomys* spp.), the consequences of increased disease prevalence and intensity could dramatically alter recovery efforts.

Plague and canine distemper are the primary diseases of concern for BFF recovery, although other diseases may impact wild BFFs (West Nile virus, tularemia, heartworm; Schuler et al. 2021). Tularemia, a bacterial disease caused by *Francisella tularensis*, can infect localized BFF and prairie dog populations through biting ticks, exposure to infected animals, and contaminated air or water; however, some BFFs seroconvert and survive infection (Matchett et al. 2021, T. Livieri and R. Santymire, unpublished data). In contrast, prairie dogs rarely seroconvert and suffer high mortality rates (Cherry et al. 2019). Unlike plague, tularemia does not cause colony collapse in prairie dogs (Tripp et al. 2017), but can cause localized mortality within a colony. That mortality also spills over into other small mammals (e.g., mice, rabbits), which can contribute to BFF diets (Brickner et al. 2014). Tularemia also is a reportable zoonotic disease of human importance and incidence of disease has increased 60% in the USA since 2011 (Bishop et al. 2023).

The BFF recovery program focuses the vast majority of disease mitigation, surveillance, and research efforts on plague because it is the most severe biological threat to in-situ populations (United States Fish and Wildlife Service 2019). Plague has affected nearly every BFF reintroduction site across the range and caused severe declines (Livieri et al. 2022) but we have also found tularemia seropositive BFFs at 6 of 12 black-footed ferret recovery sites surveyed for diseases from 2002-22 (T. Livieri and R. Santymire, unpublished data). In 2015-16, we found 37/79 (47%) of BFFs were tularemia seropositive at Conata Basin/Badlands National Park, South Dakota, site of the largest wild population. An adult male at that site had a throat mass that contained exudate with *Francisella tularensis* (see Appendix Photo 1) and while the animal otherwise appeared healthy and was administered antibiotics by our veterinarian, it was not observed on subsequent BFF surveys. A cursory look at our 2015-16 data found 52% of seronegative and 43% of seropositive BFFs were observed in subsequent surveys. These pilot data are not conclusive but provide a basis for further investigation into tularemia seroprevalence in BFFs and potential effects on BFFs at Conata Basin/Badlands.

At Conata Basin/Badlands we annually survey the BFF population to estimate population size, administer vaccines, determine demography, and collect samples for approved research projects. Notably, most BFFs are parasitized by two tick species, *Ixodes sculptus* and *I. kingi* (Harris et al. 2014), with tick prevalence on BFFs ranging from 68-100% in our surveys during 2006-22 (T. Livieri, unpublished data). Ticks are a common vector of tularemia and may be a reservoir of the bacteria (Zellner and Huntley 2019), yet very little is known about the vectors and reservoirs of tularemia in the BFF-prairie dog system. We propose to investigate ticks that parasitize BFFs and those sampled from the ecosystem to understand tularemia prevalence within ticks and elucidate potential reservoirs of this disease.

We will utilize a significant quantity of archived samples and collect additional samples from the largest wild BFF population to better understand the prevalence, reservoirs, and effects of this disease on a BFF population. The results of this study will be directly applicable to field conservation of BFFs across the range where tularemia has been documented and directly inform strategies for surveillance, sampling, and potential mitigation. BFFs are a SAFE species with a 3-year program plan (2019-21) that is currently in revision (P. Marinari, pers. comm.). The prior plan included a goal to increase AZA funding for recovery efforts in the field. In our annual field activities, we invite three AZA-accredited zoos (Dakota Zoo, Roosevelt Park Zoo, and NEW Zoo) to participate. Each of these zoos also has an active BFF display animal and

thus keepers/educators get primary field conservation experience. The US Fish and Wildlife Service recovery plan lists six action items and includes “Reduce disease-related threats in wild populations of ferrets and associated species”.

2. List the goal(s) and associated objective(s) of the proposed project and the actions that will be taken to achieve them. Goals should describe the desired status in the long-term of the question or problem that the project will address. Objectives should state the immediate results that are necessary to meet the long-term goal(s). Actions should be the specific activities the project will undertake to complete each objective. If the proposal is a request to fund part of a larger project, this section should focus on the part for which CGF funding is being requested during the one-year duration, while also providing some context regarding the larger effort. **LIMIT 4,500 CHARACTERS with spaces**

Goal 1: Assess the seroprevalence of tularemia in wild BFFs at Conata Basin/Badlands National Park, South Dakota from 2002-present.

1. Objective - Identify variables that explain tularemia exposure of BFFs including sex, age, year, and habitat characteristics in each BFF territory.

a. Action - Use archived BFF sera collected by Prairie Wildlife Research and stored at Georgia State University (GSU) in -80F freezers to determine serostatus of BFFs. We will aliquot 200-500µL per archived sample to be sent to the Wyoming State Vet Lab (WSVL) for antibody microagglutination inhibition serology analysis. This well-established test has been used by Prairie Wildlife Research and partners during BFF Biomedical Surveys (2002-2022).

b. Action - We will collect 1-2mL of blood from anesthetized wild BFFs at Conata Basin/Badlands National Park, South Dakota in summer/fall 2023-24. Blood will be collected by Travis Livieri, who has extensive experience in field blood collection, into serum separator tubes, and spun down in a centrifuge to separate the serum. Sera will be immediately frozen and sent to WSVL for analysis. Any leftover sera will be stored at National Black-Footed Ferret Conservation Center in Colorado for future studies.

c. Action - Use GIS to map locations of seropositive and seronegative BFFs from 2002-present, document spatial patterns, and identify possible differences in exposure given vegetation and soil characteristics in territories.

d. Action - Run multivariate statistical models (e.g., logistic regression, GLM) to identify relationships between variables (sex, age, year, weather) and BFF exposure to tularemia.

Goal 2: Identify locations of questing ticks in the environment, within prairie dog burrows and on the surface of prairie dog colonies and sample ticks parasitizing BFFs.

2. Objective - Locate the potential sources of questing ticks that may parasitize BFFs and sample ticks parasitizing BFFs.

a. Action - Sample ticks from the environment using CO2 traps, dragging with a flag, and burrow swabbing.

b. Action - Sample ticks from anesthetized BFFs using a fine tweezers.

c. Action - Utilize ticks sampled from BFFs by Prairie Wildlife Research (2005-present)

d. Action - Store ticks in 100% ethanol, identify to species/stage, and compare abundance of ticks collected by field sample technique using multivariate regression models.

Goal 3: Understand the role of ticks in the epizootiology of tularemia in the prairie ecosystem.

3. Objective - Identify probable reservoirs/transmitters of tularemia using DNA analysis on ectoparasites

a. Action - Use Polymerase Chain Reaction (PCR) methods to estimate prevalence of tularemia in ticks collected in the environment (2023-24) and from BFFs from 2005-present

b. Action - Use multivariate statistical models to understand the abiotic factors associated with positive tularemia ticks

c. Action - Compare PCR results to seropositive and seronegative BFFs to establish relationship between vectors and hosts

d. Action - Use Cohen's Kappa Statistic to ascertain agreement between serostatus and PCR results in ticks collected from BFFs

4. General actions for all 3 goals

a. Action - Write two manuscripts for publication consideration in the Journal of Wildlife Diseases, Journal of Parasitology, or Conservation journals.

b. Action - Hartlaub Master's thesis will be submitted to UW-Stevens Point upon graduation.

The project duration is 2.5 years and we are asking for funds to begin diagnostic analyses and ectoparasite collections. Once we purchase tick sampling equipment, we can use it for several field seasons. Hartlaub will officially begin graduate school in January 2024, but she is currently employed by Prairie Wildlife Research as a technician. Prairie Wildlife Research began field work in South Dakota in 2001, and Hartlaub worked as a technician in 2022 and will again in 2023. Our project in total will occur summer of 2023 through December 2025.

3. Explain in detail the methodology for this project and why it is appropriate. If this is a research project, clearly state the research question(s) and how it will be tested. For applied conservation projects, explain how project actions will be undertaken and why the proposed actions are appropriate for addressing the conservation issue identified. If this is a conservation education or professional training project, describe the methods as they relate to the target audience, number of people that will be reached, etc. **LIMIT 7,000 CHARACTERS with spaces**

We hypothesize that: 1) tularemia exposure of BFFs will vary by year, 2) ticks are a likely vector for transmitting tularemia in the prairie dog/burrow ecosystem, 3) ticks that are PCR positive for tularemia will be taken off of BFFs seropositive for tularemia, and 4) dragging will be the most effective method for collecting ticks from the burrow ecosystem because we will cover a larger surface area.

Goal 1: Assess the seroprevalence of tularemia in wild BFFs at Conata Basin/Badlands National Park, South Dakota from 2002-present. We will sample free-ranging BFFs inhabiting prairie dog colonies in Conata Basin (a portion of the Buffalo Gap National Grasslands administered by the US Forest Service) and Badlands National Park (administered by the National Park Service) during the late summer/fall. Prairie Wildlife Research works with both agencies to lead annual BFF surveys. We will use standardized methodology (Black-Footed Ferret Recovery Implementation Team 2016); briefly, BFFs will be located using spotlights, an unbaited live trap will be inserted into the prairie dog burrow occupied by the BFF and checked hourly. Upon capture, we will transfer the BFF from the trap into a transport tube which is labeled with capture location and observer. The captured BFF will be transported immediately to Travis Livieri at a mobile anesthesia trailer (see Appendix 1 Photo 2). Livieri will anesthetize the BFF using isoflurane, sample fleas/ticks (see Harris et al. 2014), insert a microchip (if needed), administer vaccines (F1-V plague vaccine and Purevax canine distemper), and draw 1-2mL of blood from the cranial vena cava. All of these activities will occur, with the exception of blood draw, regardless of this proposed tularemia project. Prairie Wildlife Research has a valid US Fish and Wildlife Service (FWS) Endangered Species permit (currently in renewal), FWS approved anesthesia training, and extensive experience in all methods described above. We will submit a research/IACUC proposal to FWS for blood collection and also to UW-Stevens Point.

Blood is collected into serum separator tubes, and then spun down in a centrifuge to separate the serum. The archived sera have been frozen and stored in -80°F freezers at GSU. Microagglutination inhibition analysis uses live tularemia antigen and diluted BFF sera to detect an individual's exposure to tularemia. The specific antigen will bind to antibodies in a serum sample for tularemia detection with positive titers at (>1:128). High antibody titers are indicative of recent exposure (Brown et al. 1980; Botzler and Brown 2014). Hartlaub will travel to GSU to extract 200-500µL of serum per archived sample to be sent for analysis at WSVL (Brown et al. 1980). Approximately 1,200 BFF samples from 2002-2023 will be analyzed.

Using GIS, we will map serostatus of individuals by year and identify possible trends in exposure given specific ecosystem characteristics such as soil type and vegetation. Logistic regression models will help us estimate the probability of relationships between independent variables such as sex, age, year, and seropositive BFFs. Models will provide information on tularemia hot spots and explain factors that may influence tularemia's persistence in the burrow ecosystem.

Goal 2: Identify locations of questing ticks in the environment, within prairie dog burrows and on the surface of prairie dog colonies and sample ticks parasitizing BFFs. Historically, ticks have been a predominant vector for tularemia across the US (Petersen et al. 2009). BFFs at Conata Basin/Badlands are parasitized by *Ixodes sculptus* and *I. kingi* ticks (Harris et al. 2014). Ectoparasites such as ticks spend much of their time off host (Hixson 1932) and have sensory organs on their legs used to detect a host while questing on vegetation in the environment (Salomon et al. 2020). We will begin a pilot study to determine efficacy of environmental tick sampling methods in spring/summer 2023 and continue collection in spring/summer 2024. We will employ three collection techniques for host-seeking ticks in this ecosystem: dragging with flags (Maestas et al. 2016; Salomon et al. 2020), CO2 traps (Rynkiewicz and Clay 2014), and burrow swabbing (Poje et al. 2020).

We will drag 1m-sq white flannel cloth across transects on and adjacent to active prairie dog colonies in the morning and evening during peak tick activity. We will use CO2 traps in 50 x 60m grids on active prairie dog colonies to attract questing ticks. We will place one trap 10m outside the corner of randomly selected grids, and bait with dry ice, and check them in the morning and evening. Active burrows will be swabbed by attaching a 15cm-sq white flannel cloth on a plumbing snake. The swab will be pushed as far as possible into the burrow and left inside for 30 seconds before removal. We will place swabs in zip-lock bags. Bags will be labeled with location, burrow status (active or inactive) and depth of swab (Poje et al. 2020). Ticks on host BFFs will be sampled during anesthesia as described above. Ticks will be preserved in 100% ethanol and location, date, and time will be recorded during each collection. Ticks will be identified to species, sex, and life stage using keys from the US Public Health Service (Cooley and Kohls 1945). Using multivariate regression models and ANOVA, we will compare efficacy of tick collection methods.

Goal 3: Understand the role of ticks in the epizootiology of tularemia in the prairie ecosystem. PCR analysis for tularemia will be conducted at Molecular Conservation Genetics lab at USWP. We will run DNA PCR analysis on collected

specimens to determine the role ticks play in the epizootiology of tularemia within the burrow ecosystem. PCR creates copies of small slices of DNA through a denaturing and replication process (Botzler and Brown 2014). A thermal cycler is then used to heat the samples to separate DNA strands, then cooled while primers are attached to ends of the specific sequence being copied. Finally, samples are heated again and the taq polymerase synthesizes new strands of DNA. One cycle of this process creates two new double helix strands of DNA and is repeated 30 or more times (Botzler and Brown 2014). Using PCR on our collected tick samples will allow us to determine individuals that contain tularemia DNA. Hartlaub will be trained in this technique by National Wildlife Health Center, Madison, WI. After completing PCR, multivariate statistical models will be used to estimate prevalence of tularemia in the prairie ecosystem over time, determine abiotic factors related to tularemia positive ticks, and compare serostatus of BFFs and tick PCR to determine a relationship between host and vector.

4. If AZA has provided previous funding for this or a related project through its grants programs, summarize relevant findings and identify how they relate to the current proposal. **LIMIT 1,000 CHARACTERS with spaces**

N/A

5. Describe the timeline in which components of the project will be completed within the one-year project duration. Use of bulleted/numbered lists is encouraged. **LIMIT 1,500 CHARACTERS with spaces**

- Nov. - Dec. 2023: Hartlaub travels to GSU in Atlanta to aliquot archived BFF serum samples and sends them to WSVL for tularemia exposure testing.
- Jan. - May 2024: Hartlaub begins graduate school and will be on campus to begin teaching assistantship. Tularemia serology results back from WSVL. Analysis of 2002-23 BFF tularemia serology.
- Jan. 2024: Ectoparasite identification, PCR analyses begin with ectoparasites archived and collected (2005-2023)
- May - Nov. 2024: Field work in South Dakota (Livieri until Nov., Hartlaub until Aug.) – Ectoparasite collection by Hartlaub on prairie dog colonies in Conata Basin/Badlands, BFF spotlight surveys, trapping, ectoparasite sampling, and blood collection. Serum samples sent to WSVL.
- Sept. - Dec. 2024: Hartlaub on campus (teaching assistant), ectoparasite identification and final PCR analyses.
- Oct. - Dec. 2024: Data analysis (BFF serology, ectoparasite methodology comparison, PCR tularemia concordance between ectoparasites/serology). Annual progress report submitted to AZA (October) and final report to AZA (December).

6. Define the project team by identifying the personnel included for this project, their roles, qualifications, and the key skills they bring. Discuss the PI and each CI listed on the Cover Sheet and any collaborators (persons actively involved in the completion of the project). Each collaborator's institution should be listed under **Collaborating Institutions or Organizations** on the application Cover Sheet. **LIMIT 2,500 CHARACTERS with spaces**

- Travis Livieri - (PI) Executive Director and Biologist at Prairie Wildlife Research, B.S. in Wildlife and Biology (1993) and M.S. in Natural Resources-Wildlife (2007) from University of Wisconsin-Stevens Point. Worked in the field with BFFs for 28 years, expert knowledge in BFF surveys, handling, anesthesia and related activities. 34 peer-reviewed BFF publications. Mentored and/or served on graduate committees of 8 students conducting BFF research. Role is to oversee/conduct field work in South Dakota, handle/sample all BFFs, permit compliance with FWS for handling and samples (archived and actively collected), mentor Hartlaub in the field, serve on graduate committee, assist with data analysis, writing, and publication. Responsible for all reporting to AZA.

- Shelli Dubay - (CI) Professor of Wildlife at University of Wisconsin-Stevens Point since 2005. M.S. in Pathobiology from University of Wyoming in 1996 studying ocular diseases of ungulates under direction of Dr. Beth Williams (deceased). Ph.D. in Zoology and Physiology from University of Wyoming in 2000 studying nutritional ecology of northern flying squirrels and red-backed voles under direction of Drs. Greg Hayward and Carlos Martinez del Rio. Total of 34 peer-reviewed publications and advised or co-advised 9 graduate students since 2005, 4 of which received outstanding UWSP graduate student award. Teaches Wildlife Disease, Wildlife Management Techniques. Role is graduate advisor to facilitate graduate student degree progress, mentor Hartlaub in proposal /data analysis/writing, IACUC with UW-Stevens Point.

- Madisen Hartlaub - (CI) Wildlife Technician for Prairie Wildlife Research in Conata Basin/Badlands South Dakota in summer of 2022 with duties that included trapping prairie dogs and small mammals to sample for fleas and other ectoparasites. After graduation with a B.S. in Wildlife Ecology and Management from the University of Wisconsin-Stevens

Point in December 2022, continued working for Prairie Wildlife Research with duties that include lab ectoparasite identification. As an undergraduate, worked on student team to research gray squirrels and presented results at conferences. Role as Master's Degree student is to write a degree proposal for graduate committee, collect field data in South Dakota, aliquot serum samples in Georgia, identify ectoparasites, conduct PCR analysis of ectoparasites for tularemia, analyze data, and write results for reports, thesis, and publication.

7. Describe the project's monitoring and evaluation plans. Explain how the goals and objectives of the project will be met, which indicators will be used, and how those indicators will be measured. If this project has an educational or professional training component, the educational impact should be discussed (ideally including impacts on conservation-related knowledge, attitudes/affect, and behavior). Plans may be quantitative or qualitative, and should differentiate between monitoring (ensuring proposed actions are taken) and evaluation (whether the project had the desired impact on the issue or question addressed).

LIMIT 3,000 CHARACTERS with spaces

Goal 1: Assess the seroprevalence of tularemia in wild BFFs at Conata Basin/Badlands National Park, South Dakota from 2002-present.

Objective - Identify variables that explain tularemia exposure of BFFs including sex, age, year and habitat characteristics in BFF territories.

Measures of Success:

Upon completion of the project, we will have:

- Aliquoted 200-500µL of serum from 1,200 – 1,500 archived BFF sera and sent them to WSVL for tularemia antibody titers;
- Collected blood from all BFFs trapped in 2023 and 2024 for serological analyses;
- Mapped locations of seropositive and seronegative BFFs to examine relationships with habitat;
- Statistically analyzed data to determine if sex, age, year or habitat variables explain exposure of BFFs to tularemia.

Goal 2: Identify locations of questing ticks in the environment, within prairie dog burrows and on the surface of prairie dog colonies and sample ticks parasitizing BFFs.

Objective - Locate the potential sources of questing ticks that may parasitize BFFs and sample ticks parasitizing BFFs.

Measures of Success:

Upon completion of the project, we will have:

- Sampled ticks from BFFs and the environment using several field techniques;
- Statistically analyzed ticks collected by sample type (dragging, etc.) to determine the most effective way to sample ticks in prairies.

Goal 3: Understand the role of ticks in the epizootiology of tularemia in the prairie ecosystem.

Objective - Identify probable reservoirs/transmitters of tularemia using DNA analysis on ectoparasites

Measures of Success:

Upon completion of the project, we will have:

- Determined if ticks previously collected off BFFs and those collected in 2023-24 are PCR positive for tularemia;
- Compared the PCR results from ticks sampled from BFFs that are seropositive for tularemia to those that were seronegative using Cohen's Kappa statistic.

General Measures of Success:

Upon completion of the project, we will have:

- Written two manuscripts for publication consideration in the Journal of Wildlife Diseases, Journal of Parasitology, or Conservation journals from Hartlaub M.S. thesis;
- Presented results to the public, federal employees, parties listed on BFF SAFE Program Action Plan, BFF Recovery Implementation Team (in meetings and print), and professionals in a seminar hosted by the Wisconsin Center for Wildlife and at national and regional conferences;
- Written recommendations regarding how to sample ticks from prairies and how to prevent tularemia from affecting BFFs and prairie dogs. Provide recommendations to FWS to be incorporated in disease surveillance at BFF recovery sites where tularemia is present or a concern.

8. Explain and identify the means and timeline of how the key results and lessons of the project will be disseminated and shared. Key internal and external audiences should be identified. **LIMIT 1,500 CHARACTERS with spaces**

BFFs are a species of interest for the conservation community, research personnel and the public. As a result, we are committed to communicating our results to these groups. Two manuscripts from Hartlaub's thesis (graduation in December 2025) will be submitted to peer-reviewed journals. Hartlaub's thesis defense seminar will be open to the public and will be broadcast via Zoom so conservation partners can attend from all over the United States. We will advertise the defense seminar on the website for the Wisconsin Center for Wildlife. The WCW hosts a spring seminar series each year that is open to the public. We will present our results during the seminar in Spring 2025. The College of Natural Resources has over 5,000 followers on facebook and 1,000 followers on instagram. We will work with the CNR public outreach liaison to disseminate results.

Hartlaub also will present her results at National and Regional conferences such as Midwest Fish and Wildlife Conference. Hartlaub will present results to federal employees working in the area (Badlands National Park, Buffalo Gap National Grassland) and those involved in the project. Hartlaub and Livieri will present results at annual BFF meetings. We will share our results with AZA zoos that participate in the captive breeding program for BFFs along with all authors/advisors/field partners in the BFF Action Plan for the SAFE program and invite personnel with these facilities to attend the public seminars via Zoom.

9. Have all the necessary permits been obtained for the project?

a) ☒ Yes ☐ No ☐ Not Applicable

If yes, list the permits here and attach one copy of each to your Application email. Attachments should be titled **"Permit #."**

Permit #TE-064682-1 and renewal information attached including our renewal application and a recent email from the USFWS permits office with assurance. The permit is still active because we re-applied on 3/7/2022 and the US Fish and Wildlife Service has assured us that it is still valid while they update their permitting system and we have been assured that our permit will be renewed. If you have further questions about our permit then please contact John Hughes, US Fish and Wildlife Service, National Black-Footed Ferret Conservation Center, P.O. Box 190, Wellington, CO (970) 897-2730 x62229.

- b) If the necessary permits have not yet been obtained, please explain why. **Pending permits must be obtained and submitted to AZA by 15 June 2023.**

Our State of South Dakota permit is contingent upon our federal permit from USFWS. We will submit a research proposal/IACUC to the USFWS Research/IACUC Committee in May 2023 that will propose to draw blood from BFFs for this project. All other BFF activities in this project are already covered under permit. Upon research/IACUC approval from USFWS we will submit an IACUC and Biosafety protocols to UW-Stevens Point if funds are granted.

10. If research animals are to be purchased for the project, indicate and justify the destination of the animal(s) after completion of the research. **LIMIT 750 CHARACTERS with spaces**

N/A

11. If this is a multi-year project, identify the funding strategy for subsequent years. **LIMIT 750 CHARACTERS with spaces**

UWSP is providing stipend and fringe for Hartlaub for 2 years - January 1, 2024 to December 31, 2025. We are asking AZA to cover costs for all diagnostics required for the project. Prairie Wildlife Research will cover all field costs for the summer and fall field seasons in 2023-2025.

12. **Past AZA Grant Support.** If Principal or Co-Investigators listed on the Cover Sheet have received previous AZA grant support (i.e., CGF or SAFE granting program), complete the following (required).

a) Name of Investigator: Year of Award: Amount of Award:

Project Title:

Were all required reports (annual/final) submitted?

☐ Yes ☐ No ☐ Not yet due / Other (briefly explain)

b) Name of Investigator: Year of Award: Amount of Award:

Project Title:

Were all required reports (annual/final) submitted?

☐ Yes ☐ No ☐ Not yet due / Other (briefly explain)

Note: If additional awards need to be included, add their information to a separate document titled **"Past AZA-Funded Projects"** and include it as an attachment with your Application email.

2023 AZA CONSERVATION GRANTS FUND

Project Budget

BUDGET CYCLE

All funding requests from CGF must be for a maximum period of one year only. Funding decisions are announced in early September and expenses incurred prior to the award date will not be covered by CGF funds. CGF funds are to be used within one year from the project start date and forward.

DETAILED BUDGET

A detailed budget sheet is provided below and information should be separated into CGF and non-CGF costs. List under each category the budget items and method of calculation for each (e.g., 2 persons x 10 days x \$25/day), as well as probable sources of support. The following should be listed as **separate** budget items:

- airfares
- lodging
- vehicle rental (automobiles, planes, boats)
- fuel
- per diem
- equipment over \$500
- supplies, services (lab analyses, editing)
- pharmaceuticals
- shipping/postage
- printing
- personnel costs

CGF does not fund the following: Tuition, university fees, or fringe benefits associated with graduate students or other project personnel; administrative costs (including institutional overhead and submission costs for the publication of journal articles); travel unrelated to completion of the project (conference presentation expenses, travel to meetings, etc.); planning meetings that define, rather than implement, conservation goals.

CGF does not fund salaries for project personnel other than graduate students, technicians, and start-up positions established solely for the purpose of completion of the project.

Note: Recipients of CGF grants are obligated to inform AZA (CGF@aza.org) of any pending or overlapping grants received. If such grants are received, AZA's Scientific Advisory Committee may recommend that the award amount be reduced or refunded.

Note: If additional budget items need to be included, add this information in a separate document titled "**Additional Budget Items**" and include it as an attachment with your Application email.

Budget Items Form:

Budget Item	Method of Calculation (e.g., 2 persons x 10 days x \$25/day)	CGF- Requested Amount	Non-CGF Requested Amount (if applicable)	Date Expected from Non-CGF Source (if applicable and not yet received)	Name of Non-CGF Source (e.g., Grant Fund, Matching)	Total Expense (CGF+Non-CGF)
Travel to/from Stevens Point, WI to Atlanta, GA and mileage on site	1 vehicle x 2,000 miles round-trip x \$0.65/mi (GSA rate)	\$1,300.00	\$			\$1,300.00
Lodging in Atlanta	One month rent (Airbnb)	\$3,000.00	\$			\$3,000.00
Perdiem in Atlanta for Hartlaub	1 person x \$50/day x 20 days (GSA rate)	\$1,000.00	\$			\$1,000.00
Lab supplies	Tubes, pipette tips	\$	\$300.00		Matching - PWR	\$300.00
Sample shipment	Packaging, dry ice, shipping from GA to WSVL	\$200.00	\$			\$200.00
Tularemia serology analysis (WSVL)	\$6/sample x 1,500 samples	\$9,000.00	\$			\$9,000.00
Travel to/from South Dakota field site and on site Hartlaub	1 vehicle x 1,400 miles round-trip, plus 20 mi/day x 75 days, at \$0.65/mi (GSA rate)	\$1,885.00	\$			\$1,885.00
Lodging in South Dakota for field season Hartlaub	Summer 2024 - 14 weeks \$50/day	\$	\$4,900.00		Matching - PWR	\$4,900.00
Supplies for tick sampling	CO2 traps, cloth drags, burrow swabs	\$	\$500.00		Matching - PWR	\$500.00
Taq DNA PCR supplies	Supplies for PCR testing of ticks	\$2,000.00	\$			\$2,000.00
Graduate student stipend and fringe Hartlaub	Teaching assistantship and benefits	\$	\$17,662.25		Matching - UWSP	\$17,662.25
Tuition and fees	30 graduate credits at UWSP at \$562.55 per credit	\$	\$16,876.50		Matching - PWR	\$16,876.50
Graduate student field season salary Hartlaub	Summer 2024 in South Dakota, 14	\$	\$10,080.00		Matching - PWR	\$10,080.00

BUDGET JUSTIFICATION

State why each item listed in the Budget Items Form costs what it does and indicate how each item relates to the proposed scope of work. Sufficient detail should be provided to address any potential reviewer concerns with respect to cost and need. If the lifespan of equipment costing over \$500 exceeds that of the grant period, detail and justify who will assume ownership of the item. **LIMIT 2,500 CHARACTERS with spaces**

Goal 1: Assess prevalence of tularemia in wild BFF populations across time. Currently, 1,000 BFF serum samples have been taken since 2005, and serum is stored at Georgia State University in Atlanta, Georgia. Using the US General Services Administration maximums, we request travel, housing, and per diem costs for Hartlaub for 30 days to aliquot 200-500ul of serum from archived samples to be shipped to the Wyoming State Vet Lab (WSVL) for analysis at \$6 per sample for tularemia antibody tests. With additional samples collected in the field, we anticipate sending 1,500 samples for a total of \$9,000. It is 1,000 miles (one way) from Stevens Point, WI to Atlanta, GA and GSA indicates \$0.65 per mile for travel. We anticipate Hartlaub will travel 20 miles in Atlanta daily for 30 days. Total mileage for the trip will cost \$1,690. GSA Atlanta nightly lodging rate is \$163, however renting for one month is discounted making \$3,000 our rental estimate. Per diem rate for meals in Atlanta is \$74 per day, however we anticipate a discounted \$50 per day and will only be given on working days.

Travel to SD

Goal 2: Identify how BFFs acquire ticks by determining where they occur in the environment. We will manufacture flags for sampling ticks in the environment. Estimated cost for fabric and wood is \$100. We will purchase CO2 traps to sample ticks in burrows, for around \$400. These expenses will be incurred by Prairie Wildlife Research as a match.

Goal 3: Understand tularemia transmission in prairies and identify probable reservoirs/transmitters of tularemia using DNA analysis on ectoparasites. We are working with the NWHC in Madison, WI to determine if ticks taken off BFFs contain tularemia. Additional ticks will be sampled from hosts and the environment and analyzed by Hartlaub. Hartlaub will be conducting PCR analysis on collected ticks after training, so we are asking \$2,000 for PCR taq and Eppendorf tubes for analysis. Additional shipping costs to \$200 including packaging materials and postage.

AZA CGF Grant Proposal “Understanding the ecology of tularemia in endangered in-situ black-footed ferrets and the prairie dog ecosystem”

Appendix #1

Photo 1. Exudate containing *Francisella tularensis* removed from an adult male black-footed ferret by Dr. Della Garelle during a biomedical survey at Conata Basin/Badlands National Park, South Dakota in spring 2016. Note the tick attached to the neck (gray colored) and multiple fleas. Photo by Travis Livieri.



Photo 2. Prairie Wildlife Research mobile anesthesia trailer for black-footed ferret field operations. Photo by Travis Livieri



AZA CGF Grant Proposal “Understanding the ecology of tularemia in endangered in-situ black-footed ferrets and the prairie dog ecosystem”

Citations

- Bishop, A., H. H. Wang, T. G. Donaldson, E. E. Brockinton, E. Kothapalli, S. Clark, T. Vishwanath, T. Canales, K. Sreekumar, W. E. Grant, and P. D. Teel. 2023. Tularemia cases increase in the USA from 2011 through 2019. *Current Research in Parasitology and Vector-Borne Diseases* 3:100116
<https://doi.org/10.1016/j.crpvbd.2023.100116>
- Black-Footed Ferret Recovery Implementation Team. 2016. Black-footed ferret field operations manual. U. S. Fish and Wildlife Service, Wellington, Colorado. 89p.
https://www.blackfootedferret.org/uploads/1/2/7/7/127791157/usfws_2016_bff-field-operations-manual_oct2016-.pdf
- Botzler, R. G., and R. N. Brown. 2014. Foundations of wildlife diseases. University of California Press. 464p.
- Brickner, K. M., M. B. Grenier, A. E. Crosier, and J. N. Pauli. 2014. Foraging plasticity in a highly specialized carnivore, the endangered black-footed ferret. *Biological Conservation* 169:1-5. <https://doi.org/10.1016/j.biocon.2013.10.010>
- Brown, S. L., F. T. McKinney, G. C. Klein, and W. L. Jones. 1980. Evaluation of a safranin-O-stained antigen microagglutination test for *Francisella tularensis* antibodies. *Journal of Clinical Microbiology* 11:149-148.
<https://doi.org/10.1128/jcm.11.2.146-148.1980>
- Cherry, C. C., N. A. Kwit, R. E. Ohms, A. M. Hammesfahr, R. Pappert, J. M. Petersen, C. A. Nelson, and D. E. Buttke. 2019. Tularemia (*Francisella tularensis*) in a black-tailed prairie dog (*Cynomys ludovicianus*) colony. *Journal of Wildlife Diseases* 55:944-946. <https://doi.org/10.7589/2018-05-133>
- Cooley R. A., and G. M. Kohls. 1945. The genus *Ixodes* in North America. US Government Printing Office No .184.
- Harris, N. C., T. M. Livieri, and R. R. Dunn. 2014. Ectoparasites in black-footed ferrets (*Mustela nigripes*) from the largest reintroduced population of the Conata Basin, South Dakota, USA. *Journal of Wildlife Diseases* 50:340-343.
<https://doi.org/10.7589/2013-03-048>
- Hixson, H. 1932. The life history and habits of *Ixodes sculptus neumann* (ixodidae). *Iowa State College Journal of Science* 7:34-42.
<https://www.cabdirect.org/cabdirect/abstract/19351000007>
- Leon, A. E., D. Garelle, A. Hartwig, E. A. Falendysz, H. S. Ip, J. S. Lankton, T. N.

- Tretten, T. R. Spraker, R. Bowen, and T. E. Rocke. 2022. Immunogenicity, safety, and anti-viral efficacy of a subunit SARS-CoV-2 vaccine candidate in captive black-footed ferrets (*Mustela nigripes*) and their susceptibility to viral challenge. *Viruses* 14:2188 <https://doi.org/10.3390/v14102188>
- Livieri, T. M., S. C. Forrest, M. R. Matchett, and S. W. Breck. 2022. Conserving endangered black-footed ferrets: Biological threats, political challenges, and lessons learned. Pages 458-470 in D. DellaSalla and M. Goldstein, editors. *Imperiled: The encyclopedia of conservation*. Elsevier. <https://doi.org/10.1016/B978-0-12-821139-7.00061-1>
- Maestas, L. P., S. L. Adams, and H. B. Britten. 2016. First evidence of an established population of *Ixodes scapularis* (Acari: Ixodidae) in South Dakota. *Journal of Medical Entomology* 53:965-966. <https://doi.org/10.1093/jme/tjw038>
- Matchett, M. R., T. R. Stanley, M. F. Mccollister, D. A. Eads, J. T. Boulerice, and D. E. Biggins. 2021. Oral sylvatic plague vaccine does not adequately protect prairie dogs (*Cynomys* spp.) for endangered black-footed ferret (*Mustela nigripes*) conservation. *Vector-Borne and Zoonotic Diseases* 21:921-940. <https://doi.org/10.1089/vbz.2021.0049>
- Ostfeld, R. S. 2009. Climate change and the distribution and intensity of infectious diseases. *Ecology* 90:903-905. <https://www.jstor.org/stable/25592576>
- Petersen, J. M., P. S. Mead, and M.E. Schriefer. 2009. *Francisella tularensis*: an arthropod-borne pathogen. *Veterinary Research*. 40:07. <https://dx.doi.org/10.1051/vetres:2008045>
- Poje, J. E., T. E. Rocke, and M. D. Samuel. 2020. Impacts of environmental conditions on fleas in black-tailed prairie dog burrows. *Journal of Vector Ecology* 45:356-365. <https://doi.org/10.1111/jvec.12405>
- Rohr, J. R., and J. M. Cohen. 2020. Understanding how temperature shifts could impact infectious disease. *PLoS Biology* 18(11): e3000938 <https://doi.org/10.1371/journal.pbio.3000938>
- Rynkiewicz, E. C., and K. Clay. 2014. Tick community composition in Midwestern US habitats in relation to sampling method and environmental conditions. *Experimental and Applied Acarology* 64: 109-119. <https://doi.org/10.1007/s10493-014-9798-7>
- Salomon, J., S. A. Hamer, and A. Swei. 2020. A beginner's guide to collecting questing hard ticks (Acari: Ixodidae): a standardized tick dragging protocol. *Journal of Insect Science* 20:1-8. <https://doi.org/10.1093/jisesa/ieaa073>

- Schuler, K., M. Claymore, H. Schnitzler, E. Dubovi, T. Rocke, M. J. Perry, D. Bowman, and R. C. Abbott. 2021. Sentinel coyote pathogen survey to assess declining black-footed ferret (*Mustela nigripes*) population in South Dakota, USA. *Journal of Wildlife Diseases* 57:264-272. <https://doi.org/10.7589/JWD-D-20-00015>
- Smith, K. F., K. Acevedo-Whitehouse, and A. B. Pedersen. 2009. The role of infectious diseases in biological conservation. *Animal Conservation* 12:1-12. <https://doi.org/10.1111/j.1469-1795.2008.00228.x>
- Tompkins, D. M., S. Carver, M. E. Jones, M. Krkosek, and L. F. Skerratt. 2015. Emerging infectious diseases of wildlife: a critical perspective. *Trends in Parasitology* 31:149-159. <https://doi.org/10.1016/j.pt.2015.01.007>
- Tripp, D. W., T. E. Rocke, J. P. Runge, R. C. Abbott, and M. W. Miller. 2017. Burrow dusting or oral vaccination prevents plague-associated prairie dog colony collapse. *EcoHealth* 14:461-462. <https://doi.org/10.1007/s10393-017-1236-y>
- United States Fish and Wildlife Service. 2019. Species status assessment report for the black-footed ferret (*Mustela nigripes*). Published report, Wellington, CO. 134p. <https://ecos.fws.gov/ServCat/DownloadFile/169265>
- Zellner, B., and J. F. Huntley. 2019. Ticks and tularemia: Do we know what we don't know? *Frontiers in Cellular and Infection Microbiology* 9:146 <https://doi.org/10.3389/fcimb.2019.00146>

CURRICULUM VITAE

NAME

Travis M. Livieri

ADDRESS

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Stevens Point, WI 54481
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EDUCATION

- 2023 (expected December) Ph.D. in Zoology, Colorado State University
Dissertation: Population dynamics of black-footed ferrets in relation to sylvatic plague
- 2007 M.S. in Natural Resources, University of Wisconsin – Stevens Point
Thesis: Black-footed ferret spatial use of prairie dog colonies in South Dakota
- 1993 B.S. in Wildlife and Biology, University of Wisconsin – Stevens Point

POSITIONS

Executive Director/Wildlife Biologist, 2001-present, Prairie Wildlife Research, Stevens Point, WI

Extensive research and recovery activities for black-footed ferrets and prairie dogs across North America (SD, WY, MT, KS, CO, UT, NM, AZ, Mexico, Canada).

Administrative responsibilities include day-to-day operations, budget, accounting, grant writing and administration, communication with many constituents including grantors, federal agencies, state agencies, tribes, non-profits, private landowners, universities and the general public. Close coordination with US Fish & Wildlife Service, US Forest Service and National Park Service on management of black-footed ferrets and prairie dogs. Attend and participate in meetings regarding species management.

Field responsibilities include primary field biologist that coordinates and executes black-footed ferret field activities in Conata Basin/Badlands, South Dakota. Communicates with federal partners to organize field crews, collect data, estimate population size, translocate animals, vaccinate against disease, design research studies and identify partners/students, analyze data, write peer-reviewed publications, write reports and present results to a wide variety of audiences. Serve as a consultant and partner biologist to many other black-footed ferret projects across North America and provide many of the same biological services.

Black-Footed Ferret Biological Technician, 1995-2001, US Forest Service, National Park Service, US Geological Survey Biological Resources Division, National Biological Service.

HONORS and AWARDS

2018 – University of Wisconsin Stevens Point Distinguished Alumnus Award
2012 – The Wildlife Society Conservation Education Award
2011 – Southeast Regional Emmy Award for Informational/Instructional
2009 – Regional Partner of the Year from the US Forest Service
2009 – In Recognition of the First Canadian Reintroduction of the Black-Footed Ferret
2009 – Black-Footed Ferret Protection Achievement Award from Prairie Dog Coalition
2007 – BFF Recovery Implementation Team Award for Chair of Conservation Subcom.
2007 – USFS National Grasslands Prairie Partner Award for support of NG management
2006 – International Ferret Congress for dedication to black-footed ferret recovery
2003 – US Forest Service Certificate of Appreciation for protecting resources
1995 – National Park Service On the Spot Award for work on ferret recovery

CONTRACTS and GRANTS

Total = \$1.83 million not including \$500,000+ in small donations

PUBLICATIONS

34 peer-reviewed journal articles, 4 book chapters and multiple technical reports on black-footed ferrets, prairie dogs, badgers, burrowing owls, long-billed curlews, plague, fleas and prairie conservation.

PROFESSIONAL AFFILIATIONS and SERVICE

Member – Black-Footed Ferret Recovery Implementation Team – Executive Committee, Conservation Subcommittee, Genomics Subcommittee – 2001-present
Chair – Black-Footed Ferret Recovery Implementation Team – Conservation Subcommittee – 2004-2007
Chair – AZA Black-Footed Ferret Program Review – Reintroduction Steering Comm
Affiliate Member – Canadian Black-Footed Ferret Recovery Team – 2005-present
The Wildlife Society – 2001-2008, 2012-present
Society for Conservation Biology – 2001-2004
American Society of Mammalogists – 2011-present
Front Range Community College Natural Resources Advisory Committee – 2016-2021
Colorado State Land Board Lowry Range Advisory Committee – 2015-2021
Peer Reviewer – Western North American Naturalist, Prairie Naturalist, Wildlife Society Bulletin, Journal of Mammalogy, Vector-Borne and Zoonotic Diseases, Current Zoology, Italian Journal of Zoology, Endangered Species Research, Journal of Wildlife Diseases, Ethology Ecology & Evolution, Journal of Parasitology, South African Journal of Science, Polar Research.
Graduate student committee membership and mentoring: Cynthia Maddox (MS 2004) Central Michigan University, David Eads (MS 2009) University of Missouri, David Jachowski (MS 2007) University of Missouri, Sharon Poessel (MS 2010) Colorado State University, Payton Phillips (MS 2018) Central Michigan University, Kara White (MS 2019) University of Wisconsin-Stevens Point, Becca Windell (MS 2019) Colorado State University, Mary Rae Gibson (MS expected 2023) Central Michigan University

Shelli A. Dubay

EDUCATION:

Ph.D. Zoology and Physiology, University of Wyoming, 2000. Advisers – Dr. C. Martinez Del Rio and Dr. G. Hayward

M.S. Pathobiology, University of Wyoming, 1996. Adviser – Dr. E. Williams

B.A. Environmental Science, State University of New York, College at Purchase, 1993.

CURRENT POSITION:

Professor of Wildlife, University of Wisconsin-Stevens Point, Stevens Point, WI 54481

SELECTED PUBLICATIONS (31 total, * denotes undergraduate student, ** denotes graduate student)

- **Jaworski, J., B. Strobel, and S. Dubay. (submitted) Effect of camera deployment on nesting cranes at Necedah National Wildlife Refuge. *Canadian Journal of Zoology*.
- Houtman, M., S. Dubay, J. Riddle, M. Gibson, A. Gossett, and J. Sandberg. (Accepted). Increased neurological abnormalities in Passerines and Near Passerines through seven years of avian rehabilitation in Wisconsin. *Passenger Pigeon*.
- *Shurba, J., R. A. Cole, **M. S. Broadway, C. L. Roderick, J.D. Riddle, S.A. Dubay, and S. Hull. 2021. Gapeworm (*Syngamus* spp.) prevalence in Wisconsin greater prairie chickens (*Tympanuchus cupido pinnatus*). *Journal of Parasitology*. 107:600-605. <https://doi.org/10.1645/19-138>
- Drake, D. S. Dubay, and M. L. Allen. 2021. Evaluating human-coyote interactions in an urban landscape using citizen science. *Journal of Urban Ecology* 7: 1-7. <https://doi.org/10.1093/jue/juaa032>
- *Pluemer, M., S. Dubay, D. Drake, S. Crimmins, *T. Veverka, H. Hovanec, M. Torkelson, and **M. Mueller. 2019. Red foxes (*Vulpes vulpes*) and coyotes (*Canis latrans*) in an urban landscape: prevalence and risk factors for disease. *Journal of Urban Ecology* 5:1-9. doi: 10.1093/jue/juz022
- **Zweep, J., C.N. Jacques, S. E. Jenkins, R. W. Klaver, and S. A. Dubay. 2018. Nest tree use by southern flying squirrels in fragmented midwestern landscapes. *Wildlife Society Bulletin* DOI: 10.1002/wsb.901
- Jacques, C. N., J. A. Jenks, R. W. Klaver, and S. A. Dubay. 2017. Associations among habitat characteristics and meningeal worm prevalence in eastern South Dakota. *Journal of Wildlife Diseases* 53: 131-135.
- **McKinney, L., J. Barzen, S. Dubay, J. Riddle, and T. Ginnett. 2016. Differential detection of territorial and non-territorial Greater Sandhill cranes in summer. *Proceedings of the North American Crane Workshop*. In press
- Dubay, S. A., C. N. Jacques, *N. Golden, *B. Kern, **K. Mahoney, T. R. VanDeelen, and D. Patnayak. 2015. Environmental factors that influence exposure of white-tailed deer (*Odocoileus virginianus*) to livestock pathogens in Wisconsin. *PLOS One* 10: e0128827.
- **Kamps, A., S. Dubay, J. Langenberg, and R. Maes. 2015. Efficacy of Nobuto strips for distemper and parvovirus serosurveillance in coyotes (*Canis latrans*) and raccoons (*Procyon lotor*) in southern Wisconsin (USA). *Journal of Wildlife Diseases* 51: 724-728.
- Jacques, C. N., J. A. Jenks, T. W. Grovenburg, R. W. Klaver, and S. A. Dubay. 2015. Influence of ecological factors on prevalence of meningeal worm in South Dakota. *Journal of Wildlife Diseases*. 51: 332-340.
- Dubay, S. A., *M. Buchholz, *R. Lisiecki, T. Huspeni, T. Ginnett, *L. Haen, and *P. Borsdorf. 2014. Prevalence and intensity of nematode parasites in ermine in Wisconsin. *Journal of Parasitology* 100: 616-622.
- *Samson, A. S. Dubay, T. Huspeni, and A. Cyr. 2012. Influence of environmental variables on *Baylisascaris procyonis* infection in raccoons. *Journal of Parasitology* 98: 1279-1282.
- **King, J. A., S. A. Dubay, T. C. Huspeni, A. R. VanLanen, and R. W. Gerhold. 2010. Parasite infections in nestling red-shouldered hawks (*Buteo lineatus*) in northeast Wisconsin, USA. *Journal of Parasitology* 96: 525-530.
- Dubay, S.A., T.H. Noon, J.C. deVos, Jr., and R.A. Ockenfels. 2006. Serologic survey for pathogens potentially affecting pronghorn (*Antilocapra americana*) fawn recruitment in Arizona. *Journal of Wildlife Diseases* 42: 844-848.

*Pauli, J., S. A. Dubay, E. Anderson, and S. Taft. 2004. *Strongyloides robustus* and sympatric populations of *Glaucomys volans* and *G. sabrinus* in Wisconsin. *Journal of Wildlife Diseases* 40: 579-582.

GRANTS:

State of the Birds, Federal project, USFWS. Whooping crane chick monitoring, granted to S. Dubay and J. Riddle, \$42,650 in September 2018.

University of Wisconsin Consortium for Extension and Research in Agriculture and Natural Resources. Healthy co-Existence of red fox, coyotes, domestic dogs, and humans in an urban landscape, granted to D. Drake and S. Dubay, \$59,500 in April 2016.

US Fish and Wildlife Service. Effects of forced re-nesting on reproduction of a reintroduced population of Whooping Crane (*Grus americana*), granted to B. N. Strobel and S. Dubay, \$151,692 in August 2014.

University of Wisconsin Consortium for Extension and Research in Agriculture and Natural Resources. Assessing how white-tailed deer density influences disease exposure in two sites in Wisconsin, granted to S. Dubay, T. VanDeelen, and C. Jacques, \$47,245 in April 2011.

Wisconsin Department of Natural Resources. Assessing how white-tailed deer density influences disease exposure in two sites in Wisconsin, granted to S. Dubay, \$24,129 in September 2010.

Wisconsin Department of Natural Resources, Wildlife Health. Canine parvovirus and distemper seroprevalence in urban and rural carnivore populations, granted to S. Dubay, J. Langenberg, and A. Cyr, \$23,580 in June 2008.

AGENCY REPORTS

McLean, R., J. Riddle, S. Dubay, and B. Strobel. 2019. Survival of Sandhill and Whooping Crane colts in Necedah National Wildlife Refuge. Final Report Submitted to US Fish and Wildlife Service.

Williams, E. S., J. Edwards, W. Edwards, A. McGuire, S. Dubay, W. Cook, S. Anderson, and P. Jaeger. 1997. Survey of carnivores for diseases in the Conata Basin/Badlands black-footed ferret reintroduction site, 1996-1997. Report to South Dakota Game, Fish and Parks, Department of Veterinary Sciences, University of Wyoming, Laramie, Wyoming. 12 pp.

Williams, E. S., J. Edwards, S. Dubay, C. Lynn, S. Anderson, V. Welch, and W. Edwards. 1997. Survey of carnivores for diseases in the Conata Basin/ Badlands, 1995. Report to South Dakota Game, Fish and Parks, Department of Veterinary Sciences, University of Wyoming, Laramie, Wyoming. 14 pp.

Williams, E. S., J. Edwards, S. Dubay, C. Lynn, S. Anderson, V. Welch, and W. Edwards. 1995. Survey of carnivores for diseases in the Conata Basin/Badlands, 1995. Report to South Dakota Game, Fish, and Parks. Department of Veterinary Sciences, University of Wyoming, Laramie, Wyoming. 11 pp.

Williams, E. S., J. Edwards, H. Edwards, V. Welch, S. Dubay, B. Luce, and S. Anderson. 1995. Survey of coyotes and badgers for diseases in Shirley Basin, Wyoming in 1995. Pages 43-59. *in* B. Luce, B. Oakleaf, E. T. Thorne, and E. S. Williams, editors. Black-footed ferret reintroduction in Shirley Basin, Wyoming 1995 Annual Completion Report. Wyoming Game and Fish Department, Cheyenne, Wyoming 59 pp.

Williams, E. S., J. Edwards, S. Anderson, C. Lynn, V. Welch, W. Edwards, and S. Dubay. 1994. Survey of carnivores for diseases in the Conata Basin/Badlands, South Dakota black-footed ferret reintroduction area, 1994. Report to South Dakota Game, Fish, and Parks. Department of Veterinary Sciences, University of Wyoming, Laramie, Wyoming. 21 pp.

Madisen Hartlaub

EDUCATION:

University of Wisconsin-Stevens Point (UWSP), Stevens Point, WI
Bachelor of Science Graduation: Dec. 2022
Major: **Wildlife Ecology and Management**
Minors: **Captive Wildlife** and **Biology**
Cumulative GPA: 3.64/4.00

RELEVANT EXPERIENCE:

Wildlife Technician

Prairie Wildlife Research, Stevens Point, Wisconsin/Wall, South Dakota

Jan. 2023 -present

- Organize trail camera data to determine efficacy of black-footed ferrets (BFF; *Mustela nigripes*) detection
- Mount ear mite samples onto microscope slides to determine prevalence in wild BFFs
- Identification of species, sex, and life stage and organization of flea, tick, and mite samples from BFFs

Biological Science Technician: Mammal Conservation and Plague Ecology (GS-0404-5, Step 1 equivalent)

Prairie Wildlife Research/United States Geological Survey, Wall, South Dakota

June 2022-Sept. 2022

- Monitor insecticide efficacy and plague (*Yersinia pestis*) disease ecology for endangered carnivore conservation and prairie keystone species management in collaboration with the National Park Service and the United States Forest Service
- Assess ectoparasites of black-tailed prairie dogs (*Cynomys ludovicianus*) and small mammal species
- Administer drugs for wildlife immobilization
- **Volunteer** to spotlight BFFs and observe sample collecting, PIT tagging, and vaccinations

Gray Squirrel Undergraduate Research Project Primary Co-Leader

The Wildlife Society, UWSP Student Chapter

Jan. 2020-Dec. 2022

"Investigating population dynamics and the effect of environmental variables on trap success and body condition of eastern gray squirrels (Sciurus carolinensis)"

- Trap and handle gray squirrels through ear tagging and mark recapture techniques
- Manage project budget, procure project materials and supplies, and team field schedule
- Determine ethical animal research techniques and IACUC protocol with advisor assistance

Co-Coordinator/Peer Mentor

College of Natural Resources Student Success Center (CNR SSC), UWSP

Feb. 2021-Dec.2022

Madisen Hartlaub

- Assist students regarding natural resource majors, career options, scheduling/course selection, resume and cover letter writing, student organizations, and faculty interaction.

Animal Welfare Intern

Racine Zoo, Racine, WI

May 2021-Aug. 2021

- Worked alongside animal care specialists in aspects of daily husbandry, diet preparation, exhibit cleaning and maintenance, animal observations, and behavioral enrichment for all animal routines at an AZA accredited facility
- Follow biosecurity protocol and PPE use to prevent/manage zoonotic disease

WDNR Volunteer

Wisconsin Department of Natural Resources (WDNR), Wisconsin Rapids and Deforest, WI

- Removed retropharyngeal lymph nodes to assist in monitoring for chronic wasting disease in white-tailed deer (*Odocoileus virginianus*)
Oct. 2021-Dec. 2021
- Removed and collected tooth and tissue samples of harvested bobcats (*Lynx rufus*) to assist in state furbearer species management
April 2022
- Prepared and packaged samples for submission to testing laboratory

PROFESSIONAL PRESENTATIONS:

- **Hartlaub, M.,** Luoma, N., Dubay, Dr. S., and Dr. B., Sedinger. 2022. Preliminary Study of Annual Survival of Urban Eastern Gray Squirrels (*Sciurus carolinensis*) in Schmeeckle Reserve, Stevens Point, Wisconsin. The 29th annual Conference of The Wildlife Society, Student Poster.
- **Hartlaub, M.,** Sterken, S., Dubay, Dr. S., and Dr. B., Sedinger. 2022. Influence of Calorically Dense Bait on Urban Eastern Gray Squirrel (*Sciurus carolinensis*) Body Condition in Schmeeckle Reserve, Stevens Point, WI. The 22nd annual Jim and Katie Krause CNR Student Research Symposium, Poster.
- **Hartlaub, M.,** Andexler, N., and N., Luoma. 2022. Population Estimate and Effects of Habitat and Environmental Variables on Urban Eastern Gray Squirrels (*Sciurus carolinensis*) in Schmeeckle Reserve, Stevens Point, WI. The 82nd annual Midwest Fish and Wildlife Conference, Oral.
- **Hartlaub, M.,** Funk, R., Woelfel, K., Luoma, N., Tess, A., Andexler, N., Dubay, Dr. S., and Dr. B., Sedinger. 2021. Population Estimate of Urban Eastern Gray (*Sciurus carolinensis*) Squirrels in Schmeeckle Reserve, WI. The 22nd annual Jim and Katie Krause CNR Student Research Symposium, Virtual Poster.

AWARDS:

- University of Wisconsin-Stevens Point Chancellors Leadership Award Recipient
- UWSP Student Chapter of the Wildlife Society 2022 Member of the Year
- 2020 Jim and Katie Krause CNR Student Research Symposium Highest Honors Poster Presentation



Issuing Office:

Department of the Interior
U.S. FISH & WILDLIFE SERVICE
Endangered Species Permit Office
Denver Federal Center
P.O.Box 25486
Denver, CO 80225-0489
permitsR6ES@fws.gov

MARJORIE
NELSON

Digitally signed by MARJORIE
NELSON
Date: 2017.07.17 16:49:00 -06'00'

Chief, Division of Ecological Services

Permittee:

PRAIRE WILDLIFE RESEARCH, INC.
P.O. BOX 308
WELLINGTON, CO 80549
U.S.A.

Name and Title of Principal Officer:

TRAVIS M. LIVIERI - EXECUTIVE DIRECTOR

Authority: Statutes and Regulations: 16 USC 1539(a); 50 CFR 17.22, 50 CFR 13.

Location where authorized activity may be conducted:

ON LANDS SPECIFIED WITHIN THE ATTACHED SPECIAL TERMS AND CONDITIONS

Reporting requirements:

ANNUAL REPORT DUE: 01/31
See permit conditions for reporting requirements

Conditions and Authorizations:

- A. General conditions set out in Subpart B of 50 CFR 13, and specific conditions contained in Federal regulations cited above, are hereby made a part of this permit. All activities authorized herein must be carried out in accord with and for the purposes described in the application submitted. Continued validity, or renewal of this permit is subject to complete and timely compliance with all applicable conditions, including the filing of all required information and reports.
- B. The validity of this permit is also conditioned upon strict observance of all applicable foreign, state, local tribal or other federal law.
- C. Valid for use by the permittee named above.
- D. Further conditions of authorization are contained in the attached Special Terms and Conditions.

SPECIAL TERMS AND CONDITIONS
TE064682-1
Prairie Wildlife Research, Inc.

E. Take Authorization

1. The permittee is authorized to take¹ the following federally listed species (fish and wildlife species²). Authorizations are in conjunction with the following activities for scientific/research purposes to enhance the recovery, survival, and propagation of the species in accordance with the Special Terms and Conditions stated herein.

Species Common (Scientific) Name	Status	Geographic Localities	Authorized Activities
Black-footed ferret (<i>Mustela nigripes</i>)	Endangered	Black-footed ferret reintroduction sites in the states of Arizona, Colorado, Kansas, Montana, New Mexico, South Dakota, Utah, and Wyoming.	Harass by spotlight survey/monitor, live trap, anesthetize, vaccinate, dye-mark, collect blood samples, PIT tag, collect genetic material as specified in Take Limitations below.

2. This permit does not allow take of listed species that are not specifically authorized under this permit. However, the U.S. Fish and Wildlife Service (Service) acknowledges that incidental take of a co-occurring listed species could potentially occur while conducting certain authorized activities. When applicable, the following conditions apply to all listed species that the permittee is not authorized to take under this permit, but which may be incidentally captured, injured, or killed:
 - a. Each individual authorized under this permit shall be knowledgeable about potentially co-occurring listed species that may occur throughout the habitats in which authorized activities are conducted and shall be observant and cautious to the extent that "take" of a co-occurring listed species is eliminated or minimized to the maximum extent practicable.
 - b. Any listed fish or wildlife species that the permittee is not authorized to take under this permit, but is incidentally captured during the course of conducting authorized activities, shall be released immediately at the point of capture.

¹ Take, as defined by the Endangered Species Act (ESA), means "to harass, harm, pursue, hunt, shoot, wound, kill, trap, capture, or collect, or to attempt to engage in any such conduct" for wildlife.

² The ESA defines "fish or wildlife" as any member of the animal kingdom, including without limitation any mammal, fish, bird (including any migratory, non-migratory, or endangered bird for which protection is also afforded by treaty or other international agreement), amphibian, reptile, mollusk, crustacean, arthropod or other invertebrate, and includes any part, product, egg, or offspring thereof, or the dead body or parts thereof.

- c. During the implementation of authorized activities, if an incidental³ injury or mortality occurs to a listed species not authorized under this permit, the permittee shall immediately cease authorized activities in the project area where the species/activities are occurring and notify the Project Leader (see attached list). The Project Leader will make a determination whether additional Special Terms and Conditions and/or restrictions shall be applied in this area for the species present. The Project Leader will inform the principal officer in writing of the determination within 30 days of the initial notification.
- d. Any incidental take of a listed species not authorized under this permit shall be documented in the annual report (see Reporting Requirements, below).

F. Geographic Areas

1. Authorized activities are restricted to the geographic areas noted in the table above under section E.1.
2. Authorization for property access shall be obtained by the permittee from the landowner or manager before entering Federal, State, Tribal, public, or privately owned lands to conduct authorized activities.

G. Authorized Individuals

1. Only individuals on the attached List of Authorized Individuals (List) are authorized to conduct activities under this permit. The List may identify special conditions or circumstances under which listed individuals can conduct authorized activities. Each named individual shall be responsible for compliance with the Special Terms and Conditions in this permit. The List, printed on Service letterhead, shall be retained with these Special Terms and Conditions.
2. To request changes to the List, the permittee shall submit a written request to the Recovery Permit Coordinator (see attached list). A written and mailed request shall be signed and dated by the permittee. An emailed request will be accepted from the permittee. Either type of request shall include the following information:
 - a. The name of each individual to be appended to the List;
 - b. Current position and employer's name for each individual;

³ Incidental take is defined by the ESA as take that is "incidental to, and not the purpose of, the carrying out of an otherwise lawful activity."

- c. The resume/qualifications statement of each individual, detailing their education, training, and experience with black-footed ferret and authorized activities in this permit, or similar species and activities, and type of activity for which authorization is being requested;
- d. The names, titles, organizations, emails, and phone numbers of a minimum of two references for each individual; and
- e. The names of any individuals to be deleted from the List.
- f. The permittee shall replace and attach the current updated version of the List to this recovery permit once it is received from the lead Field Office. This permit will be considered invalid without a current attached List.

Note: This procedure is for personnel changes to the List only. For requests to renew/amend this permit, a complete application shall be submitted to the U.S. Fish and Wildlife Service, Mountain-Prairie Region, Recovery Permit Coordinator, Ecological Services, P.O. Box 25486, Denver Federal Center, Denver, CO 80225, telephone 719-628-2670. **An application may be obtained at <http://www.fws.gov/forms/3-200-55.pdf>.**

H. General Permit Responsibilities

1. Acceptance of this permit serves as evidence that the permittee understands and agrees to abide by the "General Conditions for Native Endangered and Threatened Wildlife Species Permits," and the following regulations: 50 CFR Part 13 (general permit procedures), 50 CFR 17.22 (endangered wildlife), 50 CFR 17.32 (threatened wildlife), 50 CFR 17.62 (endangered plants), 50 CFR 17.72 (threatened plants), and/or 50 CFR 21.23 and 21.27 (migratory birds), as applicable (see 50 CFR 13 at http://www.ecfr.gov/cgi-bin/text-idx?c=ecfr&tpl=/ecfrbrowse/Title50/50cfr13_main_02.tpl and see 50 CFR 17 at <http://www.ecfr.gov/cgi-bin/retrieveECFR?gp=&SID=c257f650d18c76f0f932a04d854d7a8f&n=50y2.0.1.1.1&r=PART&ty=HTML>). In addition, the permittee shall have all other applicable Federal, Tribal, State, and/or local government permits prior to the commencement of activities authorized in this permit.
2. Only individuals on the List are approved to independently conduct activities under this permit. The permittee is responsible for ensuring that all authorized individuals comply with the Special Terms and Conditions in this permit.
3. The permittee and all authorized individuals must carry a copy of this permit, including attachments, while conducting authorized activities.

4. The Service requires that all handling of listed species shall be done in an expedient manner to minimize risk of injury and mortality to these individuals. Unless otherwise specified in this permit, captured individuals shall be released at their capture site as soon as authorized activities are completed or achieved.
5. Listed species and/or parts of listed species taken under this permit are not considered to be the property of the permittee. Additionally, they may not be sold, donated, or transferred without written approval from the Service, unless otherwise authorized in this permit.
6. If the permittee wants to continue work with listed species after the expiration date of this permit, a request for permit renewal (using Application Form 3-200-55) must be received by the Recovery Permit Coordinator (see attached list) **at least 30 days prior to the expiration date**. Meeting this requirement allows the permittee to continue currently authorized activities until your renewal application is acted upon. If this requirement is not met, this permit becomes invalid on the expiration date.
7. Any new activities, changes in activities, or work in new geographic areas with same or new listed species will require this permit to be amended. The permittee is not authorized to conduct any of these changes or additions until they have requested and have received an amended permit.
8. A permit renewal or amendment application will not be processed until all reporting requirements have been met for the current and previous calendar years (see Reporting Requirements, below).

I. BLACK-FOOTED FERRET Terms and Conditions

1. Only qualified and experienced individuals, as determined by the Field Office(s) (see attached list) shall independently perform the authorized activities described below.
2. The permittee is authorized to carry out the following activities within the geographic boundaries specified above, and the time limitation specified in the permit, in accordance with the Special Terms and Conditions stated below.

I.1. Presence/Absence Surveys

1. The permittee is authorized to spotlight survey, live trap, anesthetize, vaccinate, dye-mark, collect blood samples, implant Passive Integrated Transponder (PIT) tags, collect approximately 30 hairs, and scrape cells from inside of cheeks for genetic analysis provided such activities are contained within a site-specific work plan approved by the Black-footed Ferret Recovery Coordinator.

2. The permittee shall adhere to all provisions and conditions specified through annual Service black-footed ferret allocation processes with respect to the transportation, translocation, reintroduction, monitoring and handling of black-footed ferrets to and within approved reintroduction sites.
3. Only licensed project veterinarians or personnel who have successfully completed Service-sponsored ferret anesthesia training are authorized to anesthetize black-footed ferrets, and all vaccines administered by project veterinarians and/or authorized personnel must be approved in advance by the Black-footed Ferret Recovery Coordinator. No black-footed ferrets or biological samples from black-footed ferrets shall be released from the custody of the permittee to other parties without prior authorization from the Black-footed Ferret Recovery Coordinator.
4. No black-footed ferrets or biological samples from black-footed ferrets shall be released from the custody of the permittee to other parties without prior authorization from the Black-footed Ferret Recovery Coordinator.
5. The permittee shall not translocate any black-footed ferrets born or found in the wild to new locations or otherwise remove kits from the wild without prior approval from the Black-footed Ferret Recovery Coordinator.
6. The permittee shall not salvage or temporarily possess any black-footed ferret carcasses without approval of the Black-footed Ferret Recovery Coordinator. Disposition of carcasses will be at the direction of the Black-footed Ferret Recovery Coordinator.
7. The permittee may engage in photography of black-footed ferrets for scientific research purposes as documented in the permittees' research proposal in a manner which minimizes harassment to ferrets.

I.2. Take Limitations: Injury/Mortality

Species Common (Scientific) Name	Incidental Take	Intentional Take Authorized
Black-footed ferret (<i>Mustela nigripes</i>)	Incidental take in the form of harm and harassment is anticipated during the course of reintroduction events, surveys, monitoring, anesthesia, tissue sampling, and during capture and tagging and is expected to be non-lethal in nature (0 mortality). Mortality may occur from anesthesia or handling stress. These instances are expected to be rare and are not anticipated to exceed one (1) ferret over the term of the permit. In the event of any mortality, the Project Leader and the Resident Agent in Charge must be contacted within 24 hours.	Intentional mortality is not authorized.

I.5. Disposition of Mortalities

1. Any black-footed ferret that is incidentally killed or dies during covered activities shall be preserved according to standard museum practices for that species.
2. Disposition of mortalities will be at the direction of the Black-footed Ferret Recovery Coordinator.

J. Reporting Requirements

1. An annual report of activities conducted under this permit shall be submitted to the Recovery Permit Coordinators and the Project Leaders (see attached list) by **January 31** following each calendar year this permit is in effect. In order to track, document, and assess all project-specific activities conducted pursuant to this permit, we are requiring the annual report to summarize all of the activities conducted pursuant to this permit during the previous calendar year. Activities that are continuous (*i.e.*, overlapping in 2 or more calendar years), must be reported each year the activity is in effect. Example for multiple Field Offices - Each Field Office specified in the attached List shall receive separate, independent report specifying only those authorized activities conducted within their respective jurisdictions. The annual report shall be in the following format:
 - a. An introduction section addressing reasons and objectives for taking the species, as appropriate;
 - b. A methods section addressing data collection and analysis procedures, personnel working on the project, and effectiveness of the Special Terms and Conditions in minimizing take of the species;
 - c. A results section that summarizes the data collected, including information on any other federally listed species detected while conducting activities authorized under this permit; and
 - d. A conclusion section that specifically provides, at a minimum, application of the results to recommendations for the recovery of the species.
2. The annual report shall include, but need not be limited to, the following information:
 - a. Summary presentation and brief discussion of significant research results and their importance with regards to recovery of the species;
 - b. Maps and/or descriptions of locations (including GPS/GIS data, as appropriate) where permitted activities occurred;

- c. The results of all survey or sampling efforts, including estimates of population size of any federally listed species, if possible;
 - d. Number of black-footed ferret salvaged under the recovery permit, including capture locations and their deposition;
 - e. Results of any genetic studies from bio-samples collected under this recovery permit;
 - f. Quantification of take for the species, including numbers of individuals incidentally killed (including dates, locations, and circumstances of lethal take), and an estimate of the numbers of individuals otherwise harmed or harassed;
 - g. Quantification of take for other listed species not authorized under this permit, including numbers of individuals incidentally killed (including dates, locations, and circumstances of lethal take), and an estimate of the numbers of individuals otherwise harmed or harassed;
 - h. Discovery information and documentation for any potential criminal activities that was reported to the Service's Office of Law Enforcement;
 - i. Repositories where the specimens were sent, including salvaged specimens that were found malformed, abnormal, injured, sick, or dead, and any issued diagnostic or examination reports from a repository;
 - j. Other pertinent observations made during permitted activities regarding the status or ecology of the species;
 - k. Reports or other documents that include information gathered under the authority of this permit; and
 - l. Planned future activities if authorized under this permit.
3. Copies of any other Federal and/or State permits required to perform permitted activities shall be submitted with the annual report.
4. Submission of annual reports:
- a. One copy of an annual report (electronic format preferred) is required to be submitted each calendar year. Annual reports may be submitted electronically (see attached list).
 - b. Include the following subject line on the email: **Annual Report for Recovery Permit TE064682-1.**

- c. All email attachments combined are limited to 25 megabytes in size. If electronic files exceed this size limitation, please copy them onto a DVD or CD (preferred submission) or send them as a printed document(s).
 - d. Preferred formats for annual report documents are Microsoft Word, Excel, or Access; Adobe PDF; graphic files - GIF, JPG, BMP, or TIFF; and ArcGIS spatial files - shapefiles or geo-databases. If different file formats are to be submitted, contact the Project Leader or species lead before submission to verify if they will be readable by the Service.
5. If no permitted activities occurred over the course of a calendar year, indication of such by email to the above addresses shall be submitted as an annual report.
6. Failure to comply with reporting requirements may result in non-renewal or suspension/revocation of this permit.

List of Service Contacts

Black-footed Ferret Recovery Coordinator, Black-footed Ferret Conservation Center, P.O. Box 190, Wellington, CO 80459

CO/UT Resident Agent in Charge, Law Enforcement, 9297 South Wadsworth Blvd., Littleton, Colorado 80128, telephone 720-981-2777

ND/SD Resident Agent in Charge, Law Enforcement, 3425 Miriam Avenue, Bismarck, North Dakota 58501, telephone 701-355-8525

NE/KS Resident Agent in Charge, Law Enforcement, 1712 North Nelson Dr., Unit A, Derby, Kansas 67037, telephone 316-788-4474

WY/MT Resident Agent in Charge, Law Enforcement, 2900 4th Avenue North, Suite 301, Billings, Montana 59101, telephone 406-247-7356

AZ/NM Special Agent in Charge, Law Enforcement, P.O. Box 329, Albuquerque, New Mexico 87103, telephone 505-248-7895

Recovery Permit Coordinator, Ecological Services, P.O. Box 25486, Denver Federal Center, Denver, CO 80225, telephone 719-628-2670

Annual Report Submission Information

Annual reports, due by January 31, submitted electronically to the following email address and/or mailed to the following Service office:

BFF Recovery Coordinator: pete_gober@fws.gov
Black-footed Ferret Conservation Center
P.O. Box 190
Wellington, CO 80459

Recovery Permit Coordinator, Ecological Services
Mountain-Prairie Region 6 Office: permitsR6ES@fws.gov
P.O. Box 25486, Denver Federal Center
Denver, CO 80225



United States Department of the Interior

FISH AND WILDLIFE SERVICE Mountain-Prairie Region



IN REPLY REFER TO:
FWS/R6
ES

MAILING ADDRESS:
P.O. BOX 25486, DFC
Denver, Colorado 80225-0486

STREET LOCATION:
134 Union Boulevard
Lakewood, Colorado 80228-1807

LIST OF AUTHORIZED INDIVIDUALS FOR TE064682-1 Prairie Wildlife Research, Inc.

1. Individuals authorized to independently conduct all activities under this permit:

Black-footed ferret

The following individuals are authorized to independently conduct all permitted activities under the permit for the species shown.

Garelle, Della	DVM	Prairie Wildlife Research, Inc.
Livieri, Travis	Executive Director	Prairie Wildlife Research, Inc.
Santymire, Rachel	Biologist	Prairie Wildlife Research, Inc.

Black-footed ferret

The following individual is authorized only to collect approximately 30 hairs and scrape cells from inside of cheeks for genetic analysis provided such activities are contained within a site-specific work plan approved by the Recovery Coordinator:

Swanson, Brad	Biologist	Prairie Wildlife Research, Inc.
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2. Trained assistants not named on this permit may conduct activities pursuant to this permit only under the direct, on-site supervision of an independently authorized individual listed above. "On-site supervision" is defined as having the authorized individual at a distance that would allow this individual to immediately assist a supervised individual, as needed, while they are conducting an authorized activity.
3. To request personnel changes to this List of Authorized Individuals, refer to Authorized Individuals, Section G. pages 3–4, in this recovery permit.

MARJORIE
NELSON

Digitally signed by MARJORIE
NELSON
Date: 2017.07.17 16:49:00 -06'00'

Chief, Division of Ecological Services

Date

This List of Authorized Individuals (List) is valid only if it is dated on or after the permit issuance date. This permit will be considered invalid without this List.



FEDERAL FISH AND WILDLIFE PERMIT APPLICATION FORM U.S. FISH AND WILDLIFE SERVICE



Return to: U.S. Fish and Wildlife Service (USFWS)
[click here for return addresses](#)

Type of Activity: Native Endangered and Threatened Species
Scientific, Enhancement of Propagation, or Survival (i.e.,
Purposeful Take for Recovery)

Complete Sections A or B, and C, D, and E of this application. A U.S. physical address is required in Section C, see instructions for details.
Refer to the Application Form Instructions for information on how to make your application complete and help avoid unnecessary delays.

A. Complete if applying as an individual					
1.a. Last name		1.b. First name		1.c. Middle name or initial	
1.d. Suffix					
2. Date of birth (mm/dd/yyyy)		3. Occupation		4.a. Affiliation/Doing business as (see instructions)	
				4.b. Website URL (if applicable)	
5.a. Telephone number		5.b. Alternate telephone number		6. E-mail address	
B. Complete if applying on behalf of a business, corporation, public agency, Tribe, or institution					
1.a. Name of business, agency, Tribe, or institution			1.b. Doing business as (dba)		
2. Tax identification no.		3.a. Description of business, agency, Tribe, or institution		3.b. Website URL (if applicable)	
4.a. Principal officer (P.O.) last name		4.b. P.O. first name		4.c. P.O. middle initial	
				4.d. P.O. e-mail address	
5. P.O. title			6. Primary contact name		
7.a. P.O. telephone number		7.b. Alternate phone no.		8.a. Primary contact telephone no.	
				8.b. Primary contact e-mail address	
C. All applicants MUST complete					
1.a. Physical address (U.S. Street address; Apartment #, Suite #, or Room #; no P.O. Boxes)					
1.b. City		1.c. State	1.d. Zip code/Postal code	1.e. County/Province	1.f. Country
2.a. Mailing address (if different than physical address) and name of contact person (if applicable)					
2.b. City		2.c. State	2.d. Zip code/Postal code	2.e. County/Province	2.f. Country
D. All applicants MUST complete					
1. Attach the nonrefundable application processing (check or money order), payable to the U.S. FISH AND WILDLIFE SERVICE in the amount indicated on page 3. Federal, Tribal, State, and local government agencies, and those acting on behalf of such agencies, are exempt from the processing fee – attach documentation of fee exempt status as outlined in Application Form Instructions (50 CFR 13.11(d)).					
2. Do you currently have or ever had any Federal Fish and Wildlife permits (includes named on permit or List of Authorized Individuals)?					
Yes. List the number of the most recent permit you have held, or that you are applying to renew or amend:					
No.					
Certification: I hereby certify that I have read and am familiar with the regulations contained in Title 50, Part 13 of the Code of Federal Regulations and the other applicable parts in subchapter B of Chapter I of Title 50 , and I certify that the information submitted in this application for a permit is complete and accurate to the best of my knowledge and belief. I understand that any false statement herein may subject me to the criminal penalties of 18 U.S.C. 1001.					
Original or electronic signature of individual applicant/Principal Officer (no photocopied or stamped signatures)					Date (mm/dd/yyyy)

E. ALL APPLICANTS MUST COMPLETE.

Provide the information outlined in Section E. on the following pages. Be as complete and descriptive as possible. Please do not send pages that are over 8.5" x 11," videotapes, or DVDs. See page 9 for information on the Paperwork Reduction Act, Privacy Act, and Freedom of Information Act aspects of your application.

OTHER FEDERAL, TRIBAL, STATE, OR LOCAL APPROVALS OR AUTHORIZATIONS REQUIRED TO CONDUCT YOUR REQUESTED ACTIVITY

Please be aware that there may be other requirements necessary to conduct proposed activities such as obtaining permission to work on Federal or Tribal lands, a Federal bird banding permit, a Tribal, State, county or municipal permit, etc.

Have you obtained all required Federal, Tribal, State, county, municipal or foreign government approval to conduct the activity you propose?

- ☐ **Yes.** Provide a copy of the approval(s). List the Federal agency, tribe, State, county, and/or municipality involved and type of document required. Include a copy of these documents with the application.
- ☐ **I have applied.** List the Federal agency, tribe, State, county, and/or municipality involved, date of application(s), and type of permit(s). Provide the reasons why the authorizations/permits have not been issued.
- ☐ **Not required.** The proposed activity does not require issuance of other approvals and/or authorizations.
-

APPLICATION TYPE AND PROCESSING FEES

Annual reports and any other required reports under your valid permit(s) must be on file before a permit will be considered for renewal or amendment. Check the appropriate box below for the activity that you are requesting.

- ☐ **Administrative change:** You may update your name, address, telephone number, fax number, or e-mail address in your current application package on file at any time. These changes are considered administrative changes, and an application processing fee is not required. If you wish to make an administrative change, please complete pages 1-4 and indicate the information you are updating (e.g., address, telephone number, etc.). Submit completed pages 1-4 to the appropriate Regional Office (see <https://www.fws.gov/endangered/permits/recovery-permits-contacts.html>).

Requests other than an administrative change require an application processing fee, as described below. Mark the appropriate box and enclose a check or money order payable to the *U.S. Fish and Wildlife Service* in the amount indicated. If you are **fee exempt**, attach evidence or a justification and mark this box ☐ (see section D.1.).

- ☐ **New.** \$100 permit application processing fee
- ☐ **Renewal.** \$100 permit application processing fee. If you are applying to renew a valid permit, your complete application package must be received at least **30 days** prior to the expiration of the valid permit ([50 CFR 13.22](#)) to avoid a lapse in permit coverage.
- Renew** my existing valid permit (**without changes**) using my current application on file. Permit no. _____. Provide the required information under Option 1 below.
- Renew** my existing valid permit (**with changes**). Permit no _____. Below, indicate your requested amendment(s) and provide the required information under Option 2.
- ☐ **Amendment.** \$50 permit application processing fee: An amendment to a valid permit is requested at a time other than renewal. Permit no. _____.
When the information in your current application package on file has changed, then you must apply for an amendment to your valid permit. For example, **such changes may include the additions of species to the permit and/or changes in location or activities**. Please contact the Regional Recovery Permit Contact within the U.S. Fish and Wildlife Service Region of your proposed activity for technical assistance in making this determination (<https://www.fws.gov/endangered/permits/recovery-permits-contacts.html>). Provide the required information under Option 2 below.

Please indicate the amendment(s) you are requesting:

- ☐ Add species (specify) _____
- ☐ Add new activity) _____
- ☐ Add a geographic area _____
- ☐ Change in personnel _____
- ☐ Other (specify) _____

REFERRAL OF A RECOVERY PERMITTEE'S CONTACT INFORMATION (OPTIONAL)

The U.S. Fish and Wildlife Service often receives requests for contact information Permittees who could conduct endangered and threatened species (e.g., presence/absence surveys) contract work. In accordance with our Privacy Act System of Records Notice ([Permits System, Interior, FWS-21](#)), we may release the name, business address, business email address or business telephone number of those who wish to be contacted by third parties to do commercial survey activities. Such information is not normally released under the Freedom of Information Act - unless a compelling need on the part of the general public can be cited.

Please be aware that provision of Permittee contact information does not represent an endorsement by the USFWS of any particular Permittee. A referral is provided at the discretion of each U.S. Fish and Wildlife Service Regional Office as time and workload allow.

Please indicate below your preference for the release of your contact information to third parties.

- ☐ **Yes.** The U.S. Fish and Wildlife Service may release my name, business address, business email address and/or business telephone number to third parties as a referral for endangered and threatened species contract work.
- ☐ **No.** The U.S. Fish and Wildlife Service may not release my name, business address, business email address, and/or business telephone number to third parties.

SEA TURTLES

If your application involves sea turtles, please be aware that we share jurisdiction with National Marine Fisheries Service (NMF)/National Oceanic and Atmospheric Administration (NOAA) Fisheries for **sea turtles**. We evaluate applications for permits to conduct activities impacting sea turtles on land, or when applicants are conducting activities both on land and in the marine environment, and NMFS/NOAA Fisheries evaluates applications for permits to conduct activities impacting sea turtles in the marine environment. To apply for a permit to conduct activities with sea turtles in the marine environment or other species under NMFS/NOAA Fisheries jurisdiction, please contact the NMFS via their permit web page at <https://www.fisheries.noaa.gov/permits-and-forms>.

DISQUALIFICATION FACTOR

A conviction, or entry of a plea of guilty or *nolo contendere*, for a felony violation of the Endangered Species Act, Lacey Act, Migratory Bird Treaty Act, or the Bald and Golden Eagle Protection Act disqualifies any such person from receiving or exercising the privileges of a permit, unless such disqualification has been expressly waived by the USFWS Director in response to a written petition (50 CFR 13.21(c)).

Have you or, if applying as a business, any of the owners of the business, been convicted, or entered a plea of guilty or *nolo contendere*, forfeited collateral, or are currently under charges for any violations of the Endangered Species Act, Lacey Act, Migratory Bird Treaty Act, or the Bald and Golden Eagle Protection Act?

- ☐ **No.**
- ☐ **Yes.** Provide the following (use a separate page(s) if needed to complete your response:
- a) The individual's name:
 - b) Date of charge:
 - c) Location of incident:
 - d) Court:
 - e) Action taken for each violation:

SPECIFIC RELEVANT ACTIVITY REQUIRED INFORMATION: OPTION 1

Option 1. Renew an existing valid recovery permit without changes.

If you are applying to **renew an existing valid recovery permit without changes**, sign the following statement. The individual signing Section D. on page 1 of the application must also sign the following statement. This certification language is required under 50 CFR 13.22(a).

I certify that the statements and information submitted in support of my original application for a U.S. Fish and Wildlife Service Recovery permit no. TE _____ are still current and correct and hereby request renewal of that permit without changes. I also certify that all annual reports and any additional reporting requirements have been submitted to the USFWS.

Original or electronic signature of individual applicant/Principal Officer

Please legibly write or type the Signatory's name

Date

Signing the above statement completes your renewal application. Please submit completed pages 1- 5 of this application to the Regional Office covering the location of your proposed activity (see <https://www.fws.gov/endangered/permits/recovery-permits-contacts.html>). Requests for permit renewal must be complete and received by the USFWS no later than 30 days prior to the permit expiration to ensure that your current permit remains in effect while we process your request.

SPECIFIC RELEVANT ACTIVITY REQUIRED INFORMATION: OPTION 2

Option 2. New Recovery Permit, or Renewal with Amendment, or Amendment of an Existing Permit

General permit regulations for the USFWS are found at 50 CFR 13. Regulations for Recovery permits under the Endangered Species Act (ESA) can be found at 50 CFR 17.22(a)(1) for endangered wildlife species, 50 CFR 17.32(a)(1) for threatened wildlife species, 50 CFR 17.62 for endangered plant species, and 50 CFR 17.72 for threatened plant species.

Applications for a recovery permit must provide the following specific information (relevant to the activity) in addition to the general information on the previous pages of this application form. Please attach separate pages as needed. In order to assist us in processing your application, please provide the item number (i.e., A.1.a., etc.) that corresponds to the required information before each of your responses.

A. Identify species and activity:

1. For a new Recovery Permit or Amendment of an Existing Permit:

a. Provide the common and scientific names of the species being requested for coverage in the permit and their status (endangered (E) or threatened (T)). If you need to search for the scientific name of the species, please visit www.fws.gov/endangered/?ref=topbar. If you are requesting the addition of species to an existing permit, identify the species to be added to your valid permit.

b. Provide the number, age, and sex of such species to the extent known.

c. Identify the activity(ies) sought to be authorized (i.e., presence/absence survey, nest monitoring, bird banding, etc.) for each species. If you hold a valid permit and you are not requesting changes to authorized activities, indicate “No Changes”.

d. Provide the project title and project duration (start date/completion date) along with a copy of the study proposal, project funding agreement(s), etc., if applicable.

e. If you hold a valid permit and wish to amend it to delete species and/or activities, please identify activities and/or species to be deleted from your valid permit and the reason(s) for the deletion.
2. Also, for the collection of **plants from the wild on lands under Federal jurisdiction**:

a. Describe the plant part(s), and the number(s) or other type(s) of indication of material you plan to collect (i.e., whole plant, leaves, pollen, seeds, etc.).

b. If the proposed activity involves the collection of seeds from the wild, provide information that evaluates the effects of the seed collection on the reproductive potential of the species at the collection location.

B. Identify the location of the proposed activity:

1. Provide the name of each State, county, Tribal land, and the specific location of the proposed activity site(s) below. Include a formal legal description, section/township/range information, county tax parcel number, local address, or any other identifying property designation that will precisely place the location of the proposed activity site(s) below. Because the permit is enforceable; it is *required* that you list each specific State that you wish to work in.
- | Location | |
|---|--|
| State, county, tribal land, and the specific location of the proposed activity: | |
| Location Description: | |
2. If the specific study area is known at the time of application, attach a U.S. Geological Survey map of the study area in 7.5 minute quadrangle (1:24,000) scale, or other appropriately scaled map. If you plan to conduct surveys on a contract basis in the future, these maps can be provided once the specific area is known, however, the counties in which you propose to work in must be provided at this time, or at the very least, the State(s).

3. If your request is for aquatic species, identify the aquatic system (river/lake/stream name, river mile information, and drainage basin).
4. For plant species, identify the lands under Federal jurisdiction (name, address) where the proposed activities will be conducted.

C. Describe the proposed activity:

Provide a statement justifying the permit request, including the items listed below. A copy of the pertinent research or study proposal that provides the required information should be attached if available. Attach additional separate pages as necessary.

1. Describe how the activities or proposal will help recover each species.
 - a. If there is an approved recovery plan, identify the recovery tasks by number and name, if applicable. Include any additional recovery tasks identified in a Spotlight Species Action Plan, if applicable, or in a 5-year status review of the species.
 - b. Identify or provide copies of any previous or similar research conducted on this species.
 - c. If this information exists, explain how the project will attempt to answer questions not answered by earlier research.
 - d. Explain how you will coordinate your efforts with past and ongoing research studies.
2. Describe in detail the purpose(s) and objective(s) of the activities or project.
 - a. Provide the study design, sampling methods and equipment to be used.
 - b. Identify any null hypothesis or other anticipated results from the project that will support the reasoning that the project will enhance the propagation or survival of the affected species.
 - c. Include planned disposition of specimens upon completion of project.
3. Can this activity or project result in the injury, death, or removal from the wild of any individuals of the species?
 - a. If yes, describe all that apply (i.e., injury, death, removal from the wild).
 - b. For each species, please state the maximum number of individuals that would be injured, killed, or removed from the wild: *[If applicable, please identify, based on a reasonable expectation, the number of individuals likely to be injured or killed per activity.]*
 - c. Please state what will be done to minimize the possibility of injury to or death of individuals.
 - d. If the proposed activity would cause the death of individuals from the wild or removal of individuals from the wild, describe your attempts to obtain the wildlife or plant specimens currently held in captivity/nurseries/museums, or produced in captivity. You must demonstrate conclusively that existing specimens are unavailable or your study objectives require new/additional specimens. *[Provide the identity and telephone number of each contact made in this regard.]*
4. Identify contracts and agreements held for the proposed activities (attach a copy or provide the title, funding organization name and address, date of signature, and duration of the contract).

Indicate whether full funding will be available for the completion of the proposed activity. *[If you do not hold a contract at this time, but foresee receiving one, you may apply for a permit contingent upon receiving the contract(s).]*
5. If live wildlife or plants to be covered by the permit are to be held in captivity:

[Note: Under regulations at 50 CFR 17.22(a)(3) and 17.32(a)(3), escape of wildlife held in captivity must be reported immediately to our appropriate Regional Office (see <https://www.fws.gov/endangered/permits/recovery-permits-contacts.html>).

 - a. Provide a complete description, along with photographs and/or diagrams, of the area and facilities where wildlife or plant(s) will be held and/or maintained in captivity and describe arrangements for care during transportation and maintenance. Include the name and physical address of the area and facilities. *[A separate discussion specific for each species must be provided, when applicable.]*
 - b. Provide the full name and contact information of the person(s) who will care for live specimens, and include a description of their experience in caring for these or similar species, including a resume of their experience in raising, caring for, and propagating these or similar wildlife or plants.
 - c. Provide a copy of any contract or agreement you have secured for care of any live specimens collected under this permit

request if the identified facility is not affiliated with you.

- d. List mortalities and/or injuries resulting from your activities with these or similar species in the last 2 years.
 - e. Provide an explanation of each mortality event and the procedures employed or modified to eliminate any future mortality events.
 - f. Indicate your willingness to participate in a cooperative breeding or propagation program or to contribute data to a database or studbook. Holding wildlife and plants in captivity must comply with our Policy Regarding Controlled Propagation of Species Listed under the Endangered Species Act. This policy can be found on the USFWS Endangered Species web page at www.fws.gov/endangered/laws-policies/policy-controlled-propagation.html. Briefly describe how the proposed activity will comply with this policy.
 - g. State the planned disposition of the collected and/or propagated species after termination of the project/activity.
6. If working in multiple terrestrial and/or aquatic sites, provide the steps, protocols, and methodologies you will follow to prevent the spread of invasive species, infectious disease agents, and parasitic organisms, and to decontaminate vehicles and equipment.

D. Identify the persons who will conduct the proposed activity:

- 1. Provide the full name of all individuals, *including first name, middle initial, and last name*, who you propose will conduct activities under this permit (Please note that only those individuals who will be conducting the proposed activities independently without direct, and on-site supervision of an appropriately permitted individual need be included here).
 - a. If more than one activity is included in the permit application, indicate which activity(ies) will be completed by each individual.
 - b. For each listed individual, please provide a copy of each person's resume and/or curriculum vitae, in addition to specific information on previous professional training and experience conducting the proposed activities with the requested species or similar species. Information must include: dates and locations of previous activities involving these or similar species and the name of the supervising individual(s) under which such activities were conducted, and the approximate number of each species the applicant has worked with at each site.
 - c. For each listed individual, please provide at least two reference letters indicating the name, title, organization, email address, and telephone number preferably from federally permitted persons independent of each individual's place of employment, who can verify the individual's experience with the species.

END OF APPLICATION REQUIREMENTS

APPLICATION FORM INSTRUCTIONS

The following instructions pertain to U.S. Fish and Wildlife Service (USFWS) permit applications. The General Permit Procedures in [50 CFR 13](#) address the permitting process. For simplicity, all licenses, permits, registrations, and certificates are referred to as a permit.

GENERAL INSTRUCTIONS:

- Complete all relevant questions in Sections A or B, C, D, and E.
- **An incomplete application may cause delays in processing or may be returned to the applicant. Be sure you are completing in the appropriate application form for the proposed activity.**
- Print clearly or type the required response. Illegible applications may cause delays.
- Original or electronic signature of the application is required. Faxes or copies of the original signature will not be accepted.
- Mail the original application to the address at the top of page one of the applications or, if applicable, on the attached address list.
- **Keep a copy of your completed application.**
- **Please plan ahead. Allow at least 60 days for your application to be processed; however, some applications may take longer than 90 days to process ([50 CFR 13.11](#)).**
- Applications are processed in the order in which they are received.

SECTION A OR SECTION B:

Section A. Complete if applying as an individual:

- Enter the complete name of the responsible individual who will be the permittee if a permit is issued. Enter personal information that identifies the applicant.
- If you are applying on behalf of a client, the personal information must pertain to the client, and a document evidencing power of attorney must be included with the application.
- **Affiliation or Doing business as (dba):** business, agency, organizational, Tribe, or institutional affiliation directly related to the activity requested in the application (e.g., a taxidermist is an individual whose business can directly relate to the requested activity).

Section B. Complete if applying as a business, corporation, public agency, Tribe, or institution:

- Enter the complete name of the business, agency, Tribe, or institution that will be the permittee if a permit is issued. Give a brief description of the type of business the applicant is engaged in. Provide contact phone number(s) of the business. If you are applying on behalf of a client, a document evidencing power of attorney must be included with the application.
- **Principal Officer** is the person in charge of the listed business, corporation, public agency, Tribe, or institution and who is responsible for the application and any permitted activities. Often the Principal Officer is a Director or President. The **Primary Contact** is the person at the business, corporation, public agency, Tribe, or institution who will be available to answer questions about the application or permitted activities. Often, it is the preparer of the application.

ALL APPLICANTS COMPLETE SECTION C:

- A physical U.S. address is required.
- **Mailing address** is the address to which communications from USFWS should be mailed if different from the applicant's physical address.

ALL APPLICANTS COMPLETE SECTION D:

Section D.1. Application processing fee:

- An application processing fee is required at the time of application, unless exempted under 50 CFR 13. The application processing fee is assessed to partially cover the cost of processing a request. **The fee does not guarantee the issuance of a permit, nor will fees be refunded for applications for which processing has begun.**
- **Documentation of fee exempt status is not required for applications submitted by Federal, Tribal, State, or local government agencies, but must be supplied by those applicants acting on behalf of such agencies.** Such applications must include a letter on agency letterhead and signed by the head of the unit of government for which the applicant is acting on behalf, confirming that the applicant will be carrying out the permitted activity for the agency.

Section D.2. Federal Fish and Wildlife permits:

- List the permit number of your most recently issued USFWS permit.

Section D.3. CERTIFICATION:

- **The individual identified in Section A, the principal officer named in Section B, or a person with a valid power of attorney (documentation must be included in the application) must sign and date the application using original or electronic signature.** This signature legally binds the applicant to the statement of certification. You are certifying that you have read and understand the regulations that apply to the permit. You are also certifying that all information included in the application is true to the best of your knowledge, as described under 50 CFR 13. Be sure to read the statement and re-read the application and your answers before signing.

NOTICES

PRIVACY ACT STATEMENT

Authority: The information requested is authorized by the following: the Bald and Golden Eagle Protection Act (16 U.S.C. 668), 50 CFR 22; the Endangered Species Act of 1973 (16 U.S.C. 1531-1544), 50 CFR 17; the Migratory Bird Treaty Act (16 U.S.C. 703-712), 50 CFR 21; the Marine Mammal Protection Act (16 U.S.C. 1361, et seq.), 50 CFR 18; the Wild Bird Conservation Act (16 U.S.C. 4901-4916), 50 CFR 15; the Lacey Act: Injurious Wildlife (18 U.S.C. 42), 50 CFR 16; Convention on International Trade in Endangered Species of Wild Fauna and Flora (TIAS 8249), 50 CFR 23; General Provisions, 50 CFR 10; General Permit Procedures, 50 CFR 13; and Wildlife Provisions (Import/export/transport), 50 CFR 14.

Purpose: The collection of contact information is to verify the individual has an eligible permit to conduct activities which affect protected species. This helps USFWS monitor and report on protected species and assesses the impact of permitted activities on the conservation and management of species and their habitats.

Routine Uses: The collected information may be used to verify an applicant's eligibility for a permit to conduct activities with protected species; to provide the public and the permittees with permit related information; to monitor activities under a permit; to analyze data and produce reports to monitor the use of protected species; to assess the impact of permitted activities on the conservation and management of protected species and their habitats; and to evaluate the effectiveness of the permit programs. More information about routine uses can be found in the System of Records Notice, Permits System, FWS-21.

Disclosure: Response to the information requested in this form is voluntary. However, submission of requested information is required to process applications for permits authorized under the listed authorities. Failure to provide the requested information may be sufficient cause for the U.S. Fish & Wildlife Service to deny the request.

PAPERWORK REDUCTION ACT STATEMENT

We are collecting this information subject to the Paperwork Reduction Act (44 U.S.C. 3501) to provide the U.S. Fish and Wildlife Service the information needed to decide whether or not to allow the requested use and to respond to requests made under the Freedom of Information Act and the Privacy Act of 1974. The information that you provide is voluntary; however, submission of the requested information is required to evaluate the qualifications, determine eligibility, and document permit applicants. Failure to provide all required information is sufficient cause for the U.S. Fish and Wildlife Service to deny a permit. We may not conduct or sponsor, and you are not required to respond to a collection of information, unless it displays a currently valid OMB control number. OMB has approved this collection of information and assigned OMB Control No. 1018-0094.

ESTIMATED BURDEN STATEMENT

Public reporting for this collection of information is estimated to average 3 hours per response, including the time for reviewing instructions, searching existing data sources, gathering and maintaining the data needed, and completing and reviewing the collection of information. Send comments regarding this burden estimate or any other aspect of this collection of information, including suggestions for reducing this burden, to the Service Information Clearance Officer, U.S. Fish and Wildlife Service, 5275 Leesburg Pike, MS: PRB (JAO/3W), Falls Church, VA 22041-3803, or via email at Info_Coll@fws.gov. Please do not mail your completed form to this address.

FREEDOM OF INFORMATION ACT NOTICE (FOIA)

For organizations, businesses, or individuals operating as a business (i.e., permittees not covered by the Privacy Act), we request that you identify any information that should be considered privileged and confidential business information to allow the USFWS to meet its responsibilities under FOIA. Confidential business information must be clearly marked "Business Confidential" at the top of the letter or page and each succeeding page and must be accompanied by a non-confidential summary of the confidential information. The non-confidential summary and remaining documents may be made available to the public under FOIA [43 CFR 2.23 and 43 CFR 2.24].

Travis Livieri

From: PermitsR6ES, FW6 <PermitsR6ES@fws.gov>
Sent: Thursday, February 2, 2023 12:32 PM
To: tlivieri@prairiewildlife.org
Cc: Chuven, Justin R; Hughes, John P
Subject: Re: [EXTERNAL] Annual report for ES recovery permits TE-064682-1, TE-067734, and TE-069533-1
Attachments: BFF report Summer-Fall 2022.pdf

Hi Travis,
Thanks for providing the report.

I know your waiting for your permit renewal and just wanted to let you know we haven't forgot about it. We may be making some updates to our permitting strategy for BFF and will keep you updated. In the interim rest assured that your permit is still valid. Thanks for your patience.

Robert Krijgsman
Fish and Wildlife Biologist
Regional Recovery Permit Coordinator
[Mountain-Prairie Region](#)
Lakewood, Colorado Regional Office
Phone: (303) 236-4347

Treat others as you would expect to be treated

****NEW FWS E-Permits Portal at [ePermits Public Portal](#)****

*ePermits login guidance may be found by using [this link](#).

*Please note that you may already have an ePermits account if you have existing permits

From: Travis Livieri <tlivieri@prairiewildlife.org>
Sent: Wednesday, February 1, 2023 5:11 PM
To: Hughes, John P <john_hughes@fws.gov>; Schwarz, Matt <matt_schwarz@fws.gov>
Cc: PermitsR6ES, FW6 <PermitsR6ES@fws.gov>; Childers, Eddie L. <Eddie_Childers@nps.gov>; 'Dobesh, Phillip -FS' <phillip.dobesh@usda.gov>
Subject: [EXTERNAL] Annual report for ES recovery permits TE-064682-1, TE-067734, and TE-069533-1

This email has been received from outside of DOI - Use caution before clicking on links, opening attachments, or responding.

Please find attached the annual report for black-footed ferret activities in 2022 covered under permits:
TE-064682-1 (Prairie Wildlife Research)
TE-067734 (Badlands National Park, South Dakota)
TE-069533-1 (US Forest Service, Wall Ranger District, South Dakota)

Please let me know if you have any questions or need anything further.

Thanks,
Travis

Travis M. Livieri
Executive Director
Prairie Wildlife Research
P.O. Box 643
Stevens Point, WI 54481
(970) 219-1659
tlivieri@prairiewildlife.org
www.prairiewildlife.org

2023 AZA CONSERVATION GRANTS FUND

Statement of Institutional Support

One copy of this statement must be **electronically signed** by the institutional CEO/Director using the checkbox below, and submitted as an attachment to the CGF Application email titled “**SIS Institution Name**” for *each* institution or organization that fits one or more of the following criteria:

1. **Has an employee listed as a Principal Investigator or Co-Investigator on the Application Cover Sheet.**
2. **Is listed as a collaborating institution or organization on the Application Cover Sheet.**

By signing this form, all participating institutions, whether participating as a facility or through an employee, take responsibility for: 1) completing their part of the proposed project and 2) obtaining Institutional Animal Care and Use Committee (IACUC)* or Institutional Review Board (IRB)** approvals, if necessary. All accountability for conducting responsible research lies in the hands of each participating institution, not the CGF Scientific Advisory Committee.

All *Statement of Institutional Support* forms must be received as part of the CGF Application Email.

The (*Institution Name*) Prairie Wildlife Research has reviewed the proposal titled (*Proposal Title*) Understanding the ecology of tularemia in endangered in-situ black-footed ferrets and the prairie dog ecosystem and commits itself to providing the time, resources and/or financial support as stated in the application, and for obtaining IACUC and/or IRB approvals, if necessary, should the proposal be selected for funding by AZA's Conservation Grants Fund.

Travis M. Livieri and J. S. Fay

Name

Executive Director and Board President (Livieri), Board Member (Fay)

Title

(Zoos and Aquariums must include the signature of the Institutional CEO/Director above and cannot substitute the signature of the COO, General Curator, Vice President, Department Director, or Principal Investigator. Universities, Colleges, and Federal or State Government entities may substitute the signature of a Governing Official.)

3/15/2023

Date

☒ **Checking this box serves as my electronic signature to this letter of support. I affirm that I am the person listed above and understand that this letter WILL NOT be accepted unless the box is checked.**

* Institutional Animal Care and Use Committee (IACUC, for U.S. applicants) or the equivalent oversight committee (for non-U.S. applicants) approval is desirable for any proposal, but is required for any project in which animals are to be involved in invasive procedures and/or procedures that may result in pain, harm, distress, and/or altering of behavior (this includes any time an animal in the wild is sedated or anesthetized). This generally excludes samples obtained during regularly scheduled veterinary or husbandry procedures or emergency medical procedures. IACUC (or the international equivalent thereof) approval is required for both *in situ* and *ex situ* projects. Each institution involved in a project must submit the project proposal to its IACUC (or equivalent) for review. In order to be considered complete, proposals must include the Statement of Institutional Support from each participating institution/organization as recognition of compliance with these IACUC (or equivalent) approval guidelines.

If an institution is not registered as a research facility or similar facility and therefore does not have an IACUC (or equivalent), it needs to become affiliated with a university, zoo, or other body that has an IACUC (or equivalent) in order to obtain approval.

** Any projects supported through CGF funding that involve human subjects must be conducted under the auspices of a federally registered Institutional Review Board (IRB) and be consistent with 45 CFR Part 690: Federal Policy for the Protection of Human Subjects if the work or researchers are based in the USA. For members submitting from outside the USA for projects to be completed entirely outside the USA and its territories may use a process that meets the requirements of the prevailing legislation for protection of human subjects where the principal investigator is based or in the country where data collection will occur, whichever is more stringent. We note that all protocols must still conform to the types of protections outlined in 45 CFR Part 690 irrespective of the reviewing organization.

CONSERVATION GRANTS FUND

Association of Zoos & Aquariums

2023 AZA CONSERVATION GRANTS FUND

Statement of Institutional Support

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1. **Has an employee listed as a Principal Investigator or Co-Investigator on the Application Cover Sheet.**
2. **Is listed as a collaborating institution or organization on the Application Cover Sheet.**

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All *Statement of Institutional Support* forms must be received as part of the CGF Application Email.

The (*Institution Name*) University of Wisconsin-Stevens Point has reviewed the proposal titled (*Proposal Title*) Understanding the Ecology of Tularemia (*Francisella tularensis*) in Endangered Black-footed Ferrets (*Mustela nigripes*) and the Burrow Ecosystem and commits itself to providing the time, resources and/or financial support as stated in the application, and for obtaining IACUC and/or IRB approvals, if necessary, should the proposal be selected for funding by AZA's Conservation Grants Fund.

Katherine Jore

Name

Associate Vice Chancellor

Title

(Zoos and Aquariums must include the signature of the Institutional CEO/Director above and cannot substitute the signature of the COO, General Curator, Vice President, Department Director, or Principal Investigator. Universities, Colleges, and Federal or State Government entities may substitute the signature of a Governing Official.)

3/13/2023

Date

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2023 AZA CONSERVATION GRANTS FUND

Statement of SAFE Species Program Alignment

AZA's Conservation Grants Fund awards funds to members to implement SAFE program plans and advance the species program team's priorities. Because of the granting program's alignment with SAFE species programs, coordination with SAFE Species Programs is expected.

Program leaders are asked to submit this form for proposals related to their SAFE species program. To assure program review of relevant submissions, AZA SAFE Program Leaders may also be asked to serve as expert reviewers during the application review process and may express their support for a proposal at that time.

One copy of this statement must be **electronically signed** by the SAFE Species Program Leader using the checkbox below, and submitted as an attachment to the Application email titled "**SOA SAFE Species Program Name**" for *each* SAFE species program relevant to the Application.

All *Statement of SAFE Species Program Alignment* forms must be received as part of the CGF Application Email.

I affirm that, to the best of my knowledge, the proposal titled (*Proposal Title*) "Understanding the ecology of tularemia in endangered in-situ black-footed ferrets and the prairie dog ecosystem" aligns with and advances the objectives and priorities of SAFE (*Species Program Name*) Black-footed ferret.

Paul Marinari
Name

Senior Curator
Title

Smithsonian's National Zoo and Conservation Biology Institute
Organization

Black-footed ferret
SAFE Species Program

March 13, 2023
Date

☒ **Checking this box serves as my electronic signature to this letter of support. I affirm that I am the person listed above and understand that this letter WILL NOT be accepted unless the box is checked.**